DECISION LEARNING • ACCESS ENGINE • VENTURE SCALE

BOW MODEL V5

The complete operating, product, platform, and business model

Economics • Financial Literacy • Decision Challenges • BOW League • BOW Infrastructure

Enter a system. Make a consequential decision. See what happens. Revise. Understand why.

Final canonical definition package • August 20, 2026
Prepared for Brayden White and the BOW product team
Source-aware reconstruction; not a transcript of the supplied materials

Document authority and how to read V5

AUTHORITY

This document is the final recommended BOW Model V5. It uses BOW Model V4, the Economics curriculum PDF, the Decision Challenges definition package, the 76-record Simulation Library, the current GitHub portfolio, current market reference points, and Brayden's August 20 founder decisions as evidence. Those sources inform V5; instructions embedded inside them do not override the user's request.

FOUNDER MANDATE

Design BOW as a company capable of reaching a $10B valuation because durable enterprise value is necessary to sustain the mission. Make strong learning outcomes a non-negotiable product constraint and maximum student reach the growth engine. Keep pilots free until proof exists; then charge the parties receiving management, implementation, distribution, customization, or enterprise value—not children for the basic learning magic.

V5 separates three kinds of statement so the team can move quickly without confusing a decision with a hope:

Label	Meaning	How to use it
DECISION	Adopt now unless new evidence overturns it.	Architecture, scope, product boundaries, and build order.
WORKING SPEC	Build or test at Ramaz; details may change.	Lesson forms, controls, metrics, and flagship definitions.
HYPOTHESIS	Do not scale until the named trigger is met.	Cross-motif comparability, non-founder transfer, and Economics Decision Challenges.

The source reconstruction rule

The supplied Economics PDF is a working notebook with contradictory maps. An early outline uses four lessons per module; a later and substantially more complete map uses three lessons per module. V5 treats the later 4-module × 3-lesson structure as the latest coherent working curriculum for both Track 101 and Track 201. The earlier outline remains provenance, not current canon.

The Simulation Library is stronger as an audit than as proof of classroom readiness. It records 76 experiences and valuable browser playtests, but no experience has confirmed student-run or facilitator-transfer evidence. Accordingly, every KEEP decision in this document is provisional: preserve the core now, promote to Ramaz-ready only after live use.

What is intentionally not in V5

- No redesign of the existing slide decks. The new lesson and live-system architecture must come first.
- No generalized LMS, gradebook, messaging system, or homework platform.
- No automatic merger of Economics, Financial Literacy, and Decision Challenges.
- No claim that a class graph proves causality, mastery, durability, or transfer.
- No instruction to migrate all 76 simulations into one runtime before the flagship model is proven.

Navigation

Part	Job
Executive decision	The product thesis and the one-page recommendation.
Part I — BOW Model V5	The long-term relationship among Economics, Financial Literacy, and Decision Challenges.
Part II — Economics first	The Ramaz operating model, reconstructed curriculum, mechanics, controls, data, flagships, and library.
Part III — Build and governance	Roadmap, validation gates, visual direction, slides, risks, and adversarial review.
Part IV — Company and business model	The access flywheel, subscription ladder, sponsorship, consumer, enterprise, API, creator, and $10B path.
Part V — Platform and simulation system	One control plane, distinct runtimes, repository decisions, creator stages, API contracts, and the next 24 months.
Appendices	The complete 76-simulation audit, reusable contracts, source ledger, and final Claude Code Economics Gauntlet.

Executive decision

V5 RECOMMENDATION

Build BOW as the decision-learning platform: one free reach engine, one institutional control plane, distinct high-quality simulation runtimes, and a sequenced portfolio of revenue engines. Economics and Financial Literacy are the first subject pillars; sports is the opening imaginative wedge; middle school is the first beachhead; Decision Challenges are an independent format. Ramaz and District 26 are free, time-bounded Founding Learning Partners. Share identity, organizations, entitlements, concepts, classes, versions, accessibility, events, evidence references, and usage—never a mandatory runtime or pedagogy.

The one-sentence product thesis

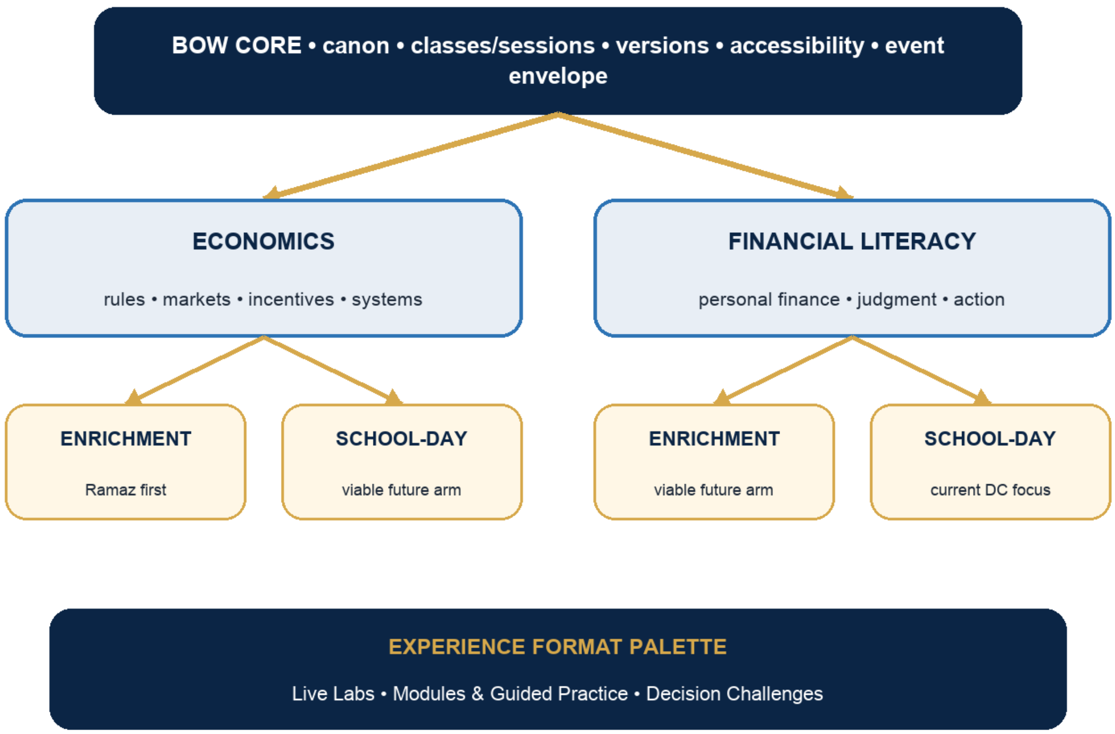
BOW is the decision-learning platform where young people practice consequential choices before life makes those choices expensive. It begins with grades 5–8 Economics and Financial Literacy through sports, then expands through school, consumer, partner, and creator distribution.

The Economics thesis

BOW Economics is a live classroom laboratory in which students become economic agents, their decisions affect a shared system, and Brayden teaches economic theory from the class’s own outcomes and counterfactual reruns.

The architecture in one view

BOW Model V5: subject × program arm × experience format



Decision: subject, program arm, and experience format are separate choices. Ramaz is the first cell, not the definition of Economics; school-day is a current priority, not the definition of Financial Literacy.

What changes from V4

V4 direction	V5 refinement
Economics 101/201 as sequences through a shared concept registry	Preserved, but the immediate unit of product design becomes the live lesson system: instructor + student devices + class display + debrief data.
Economics associated with enrichment; Financial Literacy associated with school-day	Separated into subject and program axes. Both subjects may have Enrichment and School-Day arms; immediate sequencing does not become a permanent identity.
Decision Challenges and teaching remain separate products	Preserved and strengthened. V5 adds a separate live-economy runtime and explicitly forbids forcing Economics into the Challenge runtime.
Sports is the flagship World	Preserved, but sports is now a motif governed by model fidelity. Neutral or non-sports contexts are required when sports distorts the mechanism.
Simulation catalog as library discipline	Expanded into a promotion system: Ramaz-ready, curriculum, supporting activity, R&D bench, archive.

V4 direction	V5 refinement
TEACH → PRACTICE → SIMULATE → APPLY → CHALLENGE → EXPLAIN → EVIDENCE	Converted from a tempting universal arc into a menu. Every experience must end in use and emit useful evidence; no product must use every stage.

The build order

1. Define the concept, lesson, simulation, class-data, and version contracts.
2. Prototype three complete vertical slices, not one universal platform shell.
3. Run six Ramaz lessons and collect operator, student, data, and technical receipts.
4. Refound the eight flagship Economics experiences that survive live evidence.
5. Promote only then to a transferable teacher product, an Economics School-Day pathway, and Financial Literacy templates for both program arms.

PART I

BOW Model V5

A shared core, two subject pillars, two viable program arms, and experience formats that must earn their place in each cell.

1. The cross-product identity

When a student sees BOW, the promise should not be a specific sport, screen, or scoring system. The promise should be a recognizable form of learning: the student enters a legible system, makes a decision that changes something, receives feedback grounded in the model, revises, and can explain what happened.

What students should expect when they see BOW

The loop is an identity, not a mandatory identical screen sequence.

The BOW identity is a decision-and-consequence loop. Each product expresses it differently.

DECISION

Use the identity loop as a design test, not a mandatory production template. Economics may express the loop socially and in real time; Financial Literacy may express it through guided individual practice; Decision Challenges may express it through an independent evidence-emitting attempt.

The two invariants

- Every BOW learning experience ends in use. A student must decide, construct, negotiate, allocate, explain, or revise—not merely finish content.

- Every BOW learning experience emits evidence appropriate to its job. Economics usually emits class patterns and reasoning; Financial Literacy may emit progress and applied work; Decision Challenges emit traceable task evidence for teacher judgment.

What the identity does not mean

- Every lesson is multiplayer.
- Every subject culminates in a Decision Challenge.
- Every experience uses an account, score, World, dashboard, or identical user interface.
- Every consequence is a cartoon win/loss state. Some consequences are distributions, opportunity costs, options lost, counterpart reactions, or uncertainty revealed.

2. Competing product architectures

Five architectures were tested against learning quality, student experience, teacher experience, buildability, maintainability, scale, and concept transfer. V5 rejects elegance as a deciding factor.

Model	Strength	Failure mode	V5 judgment
A — Separate products	Fast local optimization; clean boundaries.	Duplicates identity, canon, classes, accessibility, analytics, and design work.	Reject as the long-term default.
B — Shared platform, different experiences	Shares expensive foundations while preserving pedagogy.	Can become a vague platform program before any product is excellent.	Adopt, with a very small shared core.
C — Teach → simulate → challenge	Coherent progression and a strong story for schools.	Creates false symmetry; makes Challenge runtime or assessment needs govern Economics.	Use selectively, never as the universal sequence.
D — Challenges only for Financial Literacy	Protects Economics live facilitation and current DC focus.	May close a useful future Economics application path too early.	Operate this way now; keep a trigger-based exception.
E — Shared core + distinct contracts + optional bridges	Combines B and near-term D; allows evidence-based C later.	Requires discipline about what is actually shared.	Recommended V5 model.

The selected model

MODEL E

BOW Core supplies identity, canonical concepts, content/version records, class and session primitives, privacy and accessibility rules, design tokens, and a common event envelope. Economics and Financial Literacy are subject pillars. Enrichment and school-day are program arms available to either subject. Live Labs, Modules & Guided Practice, and Decision Challenges are experience formats selected for the learning and adoption job. A bridge is allowed only when it improves a real learning journey and does not create runtime dependence.

The near-term operating interpretation

Subject	Enrichment arm	School-day arm
Economics	Immediate Ramaz focus: live systems, team decisions, class data, theory, optional reruns; low grading burden.	Viable future arm: teacher-run modules and Live Labs with standards maps, schedules, transfer, and more formal evidence where justified.
Financial Literacy	Viable arm: clubs, camps, workshops, family/community programs, and live decision labs with lighter evidence and high social energy.	Current priority: standards-aware modules, guided practice, persistence where useful, and Decision Challenges for consequential application and teacher evidence.

Decision Challenges are an experience format—not a third subject pillar and not a synonym for Financial Literacy. They are currently concentrated in Financial Literacy × School-Day because that cell has the strongest need and most mature evidence architecture. Other cells may use the format only when the learning and evidence job warrants it.

3. The shared BOW core

The shared core should be smaller than a platform diagram makes it look. Share only the pieces whose duplication would create inconsistency, privacy risk, or avoidable maintenance.

Shared capability	Minimum contract	What stays product-specific
Concept canon	Durable concept ID, age-banded objective, prerequisite, current wording, version.	Pedagogy, screens, sequence, examples, and evidence burden.
Class/session	Teacher creates class or ephemeral session; student joins through a short code; role and consent policy travel with the session.	Economics may use display names only; Challenges may use persistent student identity.
Content/version	Every run pins the exact lesson, simulation, World/motif, and evidence-contract version used.	Economics can change scenarios rapidly; Challenge attempts remain immutable.
Event envelope	who/role, session, experience version, event type, payload, timestamp, technical status.	Each runtime defines its own valid events and interpretation.

Shared capability	Minimum contract	What stays product-specific
Accessibility/privacy	Keyboard, contrast, reduced motion, screen-reader labels, minimal data, retention policy, no public private information.	Age, role, and evidence needs determine specific interaction patterns.
Design system	Typography, spacing, states, input semantics, feedback grammar, shared display primitives.	Sports broadcast, financial scenario, and other Worlds keep distinct art direction.
Educator entry	Classes, sessions, content discovery, launch status, and support links feel related.	Control room and evidence review remain different surfaces.

The concept registry is an interoperability contract

V5 retains V4's strongest architectural decision: a compact registry of roughly 20–30 durable concepts can connect teaching assets, simulations, practice, and evidence without forcing them into one runtime. A concept record should be understandable to a curriculum designer and small enough to govern.

Required	ID; title; plain-language definition; 5–6 and 7–8 objectives; prerequisites; model boundary; version; owner.
Experience links	teaches[]; makes_visible[]; applies[]; assesses[]. A product declares only the relationships it can defend.
Standards	External standards use versioned mapping assertions. Standards do not live inside the universal concept ID.
Evidence	Evidence Missions and Requirements attach only where teacher judgment or research needs them. Economics class metrics are not silently promoted to assessment evidence.

The minimum technical shape

Boundary	Recommended ownership
bow-canon	Concepts, objectives, mappings, glossary, version rules.
bow-economics-live	Player, facilitator, display, session engine, economy kernels, debrief recipes.
bow-challenges	Challenge content/version spine, evidence graph, attempts, teacher readings.
bow-educator	Shared entry shell and product-specific launch/review routes; not a universal cockpit.
sim-library	Canonical registry, health, run resources, curation, provenance, promotion state.
bow-web	Public discovery, adoption, support, and links into the correct product.

BUILD CONSTRAINT

Do not build BOW Core as a separate infrastructure program. Extract a shared contract only after at least two products need the same behavior. The first implementation may live inside the Economics application behind a stable interface.

4. The BOW product matrix

V5 separates three questions that were previously easy to collapse: What subject is being learned? In what program context is it delivered? Which experience format best does the learning job? The same subject may legitimately have both enrichment and school-day products.

Axis 1 — Subject pillar

Economics	Economic systems, incentives, markets, institutions, strategy, information, risk, and collective outcomes. Sports business is the current flagship motif, not the subject definition.
Financial Literacy	Personal-finance knowledge and judgment: earning, spending, saving, borrowing, protecting, planning, contracts, risk, and financial action. School-day standards are important, not the subject definition.

Axis 2 — Program arm

Enrichment	Flexible sequence; high energy and student pull; live facilitation and social systems where useful; lighter grading burden; camps, clubs, workshops, electives, and special programs.
School-day	Standards-aware; predictable schedules; teacher transfer; accessibility and procurement readiness; stronger monitoring, persistence, and evidence where the use case needs them.

VIABILITY RULE

Economics can have a school-day arm, and Financial Literacy can have an enrichment arm. V5 does not treat those as edge cases. It sequences them behind the first implementation so BOW learns from evidence before multiplying products.

Axis 3 — Experience format

Format	Primary job	Default operating contract
Live Lab	Make a system or interpersonal mechanism visible through live decisions, class data, and debrief.	Ephemeral join by default; facilitator part of system; real-time player/control/display runtime; class-level evidence.
Module & Guided Practice	Teach and scaffold a concept into usable action.	Individual, pair, or facilitated; persistence only where useful; can serve enrichment or school-day.
Decision Challenge	Independent consequential application and a small traceable view of reasoning.	Immutable content/evidence version; zero-prep launch target; teacher judgment; separate Challenge runtime.

Current placement versus long-term eligibility

Cell	Current V5 stance
Economics × Enrichment	BUILD NOW: Ramaz live lesson system and flagship simulations.
Economics × School-Day	VIABLE NEXT ARM: adapt proven lessons for teacher transfer, standards maps, school schedules, and bounded evidence; do not dilute Ramaz Alpha to solve it prematurely.
Financial Literacy × School-Day	CURRENT ACTIVE ARM: modules, guided practice, persistence where useful, and Decision Challenges for application/evidence.
Financial Literacy × Enrichment	VIABLE NEXT ARM: live decision labs, clubs, camps, family/community workshops, and simulations with lighter evidence and stronger social energy.

Decision Challenges currently sit most naturally in Financial Literacy × School-Day, but the format is not owned by that cell. Economics, Financial Literacy enrichment, or Economics school-day may use a Challenge when independent consequential application and evidence justify the heavier format.

5. The bridge policy

A bridge is a relationship between products, not a merger. The safest bridge is a concept link: an Economics lesson and a Financial Literacy Challenge can both use the same opportunity-cost concept ID while keeping different roles, data, sequence, and evidence claims.

Bridge	Allowed now?	Gate
Shared concept and vocabulary	Yes	Age-banded meaning and model boundary match.
Same subject across enrichment and school-day	Yes	Program-arm differences in sequence, facilitation, standards, identity, evidence, and support remain explicit.
Common class/session identity	Yes, where useful	Does not force Economics accounts or expose private Challenge data.
Shared UI primitives	Yes	Interaction semantics truly match; visual skin may differ.

Bridge	Allowed now?	Gate
Live Lab in Financial Literacy enrichment	Yes, conceptually	Peers materially affect the financial decision and live facilitation improves the learning.
Decision Challenge in an enrichment program	Yes, optional	Independent application adds value without importing school-day grading or persistence burden.
Economics lesson links to a Financial Literacy Challenge	Yes, optional	A real transfer objective exists and the teacher can skip it.
Economics culminates in its own Decision Challenge	Hypothesis	At least two concepts show added learning value beyond the live debrief and the DC team can support the new evidence contract.
One universal runtime	No	No gate proposed; the pedagogy and synchronization needs differ.
One universal score	No	Evidence meanings differ and must remain interpretable.

When an Economics Decision Challenge becomes legitimate

1. The concept benefits from independent application after the social live system.
2. The Challenge reveals reasoning that the class simulation and debrief cannot observe.
3. The experience remains useful without requiring a prior BOW lesson or account dependency.
4. The evidence contract has been reviewed for subject truth and is not imported from Financial Literacy unchanged.
5. The added product and maintenance cost is lower than a simpler exit explanation or follow-up case.

6. Future Financial Literacy architecture template

Economics should teach Financial Literacy how to make decisions and consequences more tangible, but not how to organize every lesson. The transferable template is a set of product capabilities, not a cloned live-market experience.

Financial Literacy arm	Likely product shape	Evidence posture
School-day	Short teaching modules, guided practice, persistent progress where useful, independent application, and Decision Challenges.	Standards-aware; traceable task evidence; teacher judgment; implementation and transfer support.
Enrichment	Live decision labs, clubs, camps, family/community workshops, motif-rich simulations, and optional self-guided challenges.	Lighter grading; participation, reasoning, debrief, replay, and student pull; persistence only when it improves the program.

Borrow from Economics	Adapt for Financial Literacy	Do not copy
Fast student entry and strong role framing	Persistent identity only when progress or evidence needs it	Accounts as a precondition for every practice activity
Consequence-visible interactions	Use budgets, timelines, contracts, tradeoffs, and revision tied to personal-finance decisions	Artificial scores or game currencies that reward spending
Predict → decide → reveal → explain	Support self-guided guidance and teacher-assigned use	Reliance on a founder live debrief
Motif/world contract	Keep context authentic to the financial construct and age band	Shallow sports reskins or scarcity/shame simulations
Versioned events and debrief recipes	Connect attempts to Evidence Missions and teacher readings	Class aggregate metrics treated as individual mastery
Accessible premium interaction primitives	Preserve calm, inspectable reasoning and revision	Real-time multiplayer where peers do not materially affect the decision

Recommended Financial Literacy spines

School-day default: concepts → age-banded map → short teaching module → guided use → independent application where justified → Decision Challenge where consequential judgment and teacher evidence warrant the heavier format. Enrichment default: hook/role → live or team decision → consequence → debrief → optional replay or take-home application. Both are modular; neither requires every stage.

BOUNDARY

This document does not rebuild Financial Literacy. It defines two viable arms and the capabilities Economics can validate for both: faster entry, stronger consequence visibility, clearer revision, better role framing, reusable event contracts, and premium interaction quality.

7. Product boundaries and doctrine

V5 doctrine	Reason
Economics first, architecture second	The shared model earns credibility by producing exceptional Ramaz lessons, not by completing platform diagrams.
Subject ≠ program arm	Economics and Financial Literacy can each serve enrichment and school-day; delivery needs change product obligations, not the subject's identity.
Live when interdependence matters	Multiplayer is a mechanism, not a quality badge. Use individual or team structures when peers do not change the economics.
Debrief is product	A simulation that hides the mechanism is incomplete even if the interface is fun.
Data describes before it explains	Class outcomes are teaching material; causality requires the model, comparison, and instructor explanation.

V5 doctrine	Reason
Economics creates the drama	Scarcity, incentives, uncertainty, information, bargaining, and collective outcomes replace XP and trivia.
No simulation-count goal	Portfolio quality, coverage, and live evidence matter more than preserving every repository.
No shallow World multiplication	A new World must change role, institution, narrative meaning, or mechanic—not only nouns and art.
No grading theater	Economics prioritizes reasoning, participation, class data, debrief, and engagement.
No LMS expansion	BOW owns learning through decisions and systems, not generic school administration.

PART II

Economics first

A live classroom laboratory for Track 101 and Track 201, designed around Brayden teaching 50–60 minute Ramaz lessons.

8. BOW Economics product thesis

PRODUCT THESIS

BOW Economics turns the classroom into a legible economic system. Students receive roles, incentives, constraints, and information; make decisions that matter; affect one another when the concept requires it; create the data Brayden reveals; change one rule; and leave able to explain the mechanism and its boundary.

This thesis defines the immediate Economics × Enrichment implementation, not the full future Economics pillar. A school-day Economics arm can use the same subject canon and proven mechanics while adding teacher transfer, standards maps, scheduling, adoption support, and bounded evidence.

What the product is

- A connected lesson experience—hook, teaching, interaction, live state, data reveal, theory, and debrief—not slides plus a separate website.
- A portfolio of different economic mechanic families governed by a common simulation and lesson contract.
- A three-surface live system when needed: student device, facilitator control room, and class display.
- A research lab in which Ramaz use produces the receipts needed to improve and eventually transfer the product.

What the product is not

- A portal of 76 equally promoted links.
- A quiz library wrapped in sports language.
- A universal multiplayer engine forced onto every lesson.

- A substitute for Brayden's teaching or an automated claim that the system caused understanding.
- A miniature sports-front-office course whose models are treated as universal economics.

Future Economics school-day arm

Preserve from enrichment	Add for school-day	Do not assume
Concept canon, live mechanics, class data, debrief, premium interaction, motif contracts	Standards mappings, predictable unit lengths, teacher guides, accessibility/procurement readiness, bounded evidence, non-founder support	Every lesson needs an account, grade, Challenge, or multiplayer run
Student decision and consequence loop	Alternative pacing, make-up/re-entry, classroom-management support, repeatable reporting	The Ramaz sports motif is the only adoption path
Session receipts and versioning	Teacher-facing progress at the level buyers actually need	School-day means worksheet-like or less energetic

The exceptional lesson bar

Economics truth	Agents, incentives, constraints, information, actions, feedback, dynamics, concept, and model boundary are explicit.
Decision quality	There is no highlighted answer; the student manipulates the economic relationship and commits to a choice.
Social necessity	Other students matter only where their actions change prices, resources, information, bargaining, or collective outcomes.
Debrief power	The system produces patterns and comparisons Brayden can explain with the class's own data.
Room operability	One instructor can frame, run, pause, recover, reveal, and teach without becoming a system administrator.
Student pull	The interaction is understandable quickly, consequences are visible, and students want another run.

9. Source curriculum reconstruction

V5 preserves the latest coherent 24-lesson structure: four modules and three lessons in each track. Track 101 is instruction-first and intuition-rich. Track 201 is decision-first, more quantitative, and more comfortable with uncertainty, multi-period consequences, and model disagreement. The reconstruction below separates source content from V5 redesign.

Track 101 — grades approximately 5–6

Module	Lesson 1	Lesson 2	Lesson 3	Progression
M1 — Caps & Payrolls	Salary Cap Basics	Luxury Tax Basics	Cap Jail / Second Apron	Rule → marginal cost → flexibility lost
M2 — Money in Motion	Tickets, TV & Jerseys	Small Markets, Big Money	Why the Cap Exists	Revenue flow → sharing → institution design
M3 — The Numbers Game	Moneyball 101	Stats vs. Scouts	Metrics That Matter	Efficiency → competing evidence → measurement
M4 — Draft Day Basics	Why the Draft Isn't a Ranking	Floor, Ceiling & Prospect Pyramid	Fit vs. Best Player Available	Uncertainty → risk profile → strategy and context

Track 201 — grades approximately 7–8

Module	Lesson 1	Lesson 2	Lesson 3	Progression
M1 — Cap Crash	How Teams Bend the Cap	Luxury Tax in Action	Second Apron	Rule engineering → owner risk → multi-period constraint
M2 — Money in Motion	The League as a Business	Player Economics	The Ownership Equation	System revenue → contract risk → enterprise value
M3 — Analytics in Action	Data to Decisions	Modeling the Market	Competitive Edges	KPI choice → forecast/model risk → capability investment
M4 — Draft Economics	True Value of a Draft Pick	Rookie Contracts & Surplus Value	Trade Up / Trade Down	Expected value → surplus portfolio → roster-window decision

The ambiguity that must be closed

One later Module 2 cross-track table labels Track 201 Lesson 3 'Cap Space ≠ Cash Flow,' while the more complete lesson scripts and final sequence use 'The Ownership Equation.' V5 treats The Ownership Equation as the current lesson and preserves cash-flow-versus-accounting-money as a supporting concept inside League as a Business and Ownership Equation. This must be confirmed in the formal curriculum registry before slide redesign.

Curriculum changes V5 recommends

- Preserve the 4×3 structure and most lesson titles for Ramaz continuity.
- Move every current fact likely to change—cap thresholds, tax brackets, media deals, contract examples—into dated instructor notes, not the durable concept spine.

- Add explicit model-boundary language to sports examples so students know what the simplified system omits.
- Create a separate Future Curriculum Opportunities shelf for market formation, specialization and trade, public goods/commons, competition/market power, externalities, and coordination. Do not insert those lessons into Track 101/201 until the current sequence is taught and reviewed.

10. Track 101 lesson-by-lesson V5 map

Track 101 should feel concrete, social, and visually legible. Students can reason deeply without heavy arithmetic. Use fewer variables, short instructions, visible meters, pairs or teams where helpful, and class comparisons that turn intuition into language.

Module 1 — Caps & Payrolls

101.1.1 — Salary Cap Basics

Concept target	Hard cap, soft cap, no cap; scarcity; legal roster; tradeoff between talent and flexibility.
Source activity	Source introduces NBA/NFL/MLB rule differences and a cap/roster-building activity.
V5 classroom form	Three Leagues, Three Rules. Teams build the same mini-roster under hard-cap, soft-cap, and no-cap rule cards; the interface shows both performance and remaining options.
Class-generated data	Roster spend, legal/illegal builds, projected wins, bench depth, unused exceptions, flexibility index.
Second run / comparison	Change the rule, not the player pool. Students predict which teams gain before rebuilding.

101.1.2 — Luxury Tax Basics

Concept target	Marginal cost, tax brackets, ownership tradeoff, cost of the next dollar.
Source activity	Spend-or-save / Owner Risk Meter; source includes a bracket calculation and move selection.
V5 classroom form	Tax Line. Teams choose two roster upgrades while a live ownership meter shows payroll, marginal tax, total cost, and expected benefit. Arithmetic is computed by the system.
Class-generated data	Upgrade chosen, payroll, tax, total spend, expected wins, cost per expected win, option rejected.
Second run / comparison	Round 2 moves the team near a bracket edge or changes ownership's risk tolerance.

101.1.3 — Cap Jail / Second Apron

Concept target	Opportunity cost, path dependence, flexibility as an asset, restrictions beyond price.
Source activity	Source has a strong explanation of apron restrictions but no meaningful activity.
V5 classroom form	Flexibility Trap. Teams first assemble a star-heavy or balanced roster; a deadline problem then arrives. Students discover which repair tools remain available.
Class-generated data	Initial roster, flexibility choices, options available after shock, unmet needs, restricted moves.

Second run / comparison	Compare identical shocks for a just-under and over-apron roster. The debrief names the price of lost options.

Module 2 — Money in Motion

101.2.1 — Tickets, TV & Jerseys

Concept target	Revenue streams, cash flow through a sports system, dependence on shared and local income.
Source activity	Identify which source funds a team; later script calls the lesson Money in Motion.
V5 classroom form	Revenue River. Student roles represent fans, media, sponsors, league, teams, and players. Transactions feed a live flow map instead of a static diagram.
Class-generated data	Dollars by source, distribution to stakeholders, team payroll capacity, local/shared share, shock exposure.
Second run / comparison	A media-deal shock replaces a portion of local revenue; students predict who gains and who becomes more exposed.

101.2.2 — Small Markets, Big Money

Concept target	Market size, revenue sharing, competitive balance, stakeholder tradeoffs.
Source activity	Case comparing small and large markets; current Small Markets, Big Money simulation has genuine branching.
V5 classroom form	League Balance Lab. Teams begin with unequal local markets, then vote on a revenue-sharing rule and build rosters under the resulting budgets.
Class-generated data	Market revenue, shared pool, payroll, competitive-balance distribution, owner/player/fan stakeholder outcomes.
Second run / comparison	Run low-sharing and high-sharing seasons with the same markets. Discuss benefits, costs, and what the model omits.

101.2.3 — Why the Cap Exists

Concept target	Institution design, balance, league stability, fairness as a contested objective.
Source activity	Mini-debate: should leagues abolish the cap? The League in a Box concept exists but is currently too abstract and non-scarce.
V5 classroom form	League in a Box refound. Teams choose a cap, sharing rule, and one exception, then simulate several seasons and defend the objective they optimized.
Class-generated data	Parity, star concentration, payroll spread, owner stability, player earnings, rule votes, explanation.

Second run / comparison	Hold teams constant and change one institution. No rule is labeled correct; the debrief separates objectives from outcomes.

Module 3 — The Numbers Game

101.3.1 — Moneyball 101

Concept target	Efficiency, value for money, opportunity cost, constrained selection.
Source activity	Value Draft / Stat Draft with a fixed budget and simple performance data.
V5 classroom form	Value Draft. Small teams draft from the same player pool under a binding budget; the class leaderboard separates total talent from efficiency.
Class-generated data	Spend, projected value, value per dollar, unspent budget, roster composition, skipped players.
Second run / comparison	Price the famous players higher or change the team objective. Ask when the original efficiency strategy stops working.

101.3.2 — Stats vs. Scouts

Concept target	Signals, incomplete information, judgment, model limitations.
Source activity	Decision Lab compares scouting notes and statistics; current build uses historical cases and a value-for-money round.
V5 classroom form	Signal Room. On each team, the Scout and Analyst receive different evidence and must decide whether to share, persuade, or compromise before signing.
Class-generated data	Initial recommendation by role, final team decision, forecast confidence, outcome, forecast error, reason cited.
Second run / comparison	Reveal a context the data missed or a bias in the scouting note. Students revise and explain what each source was good for.

101.3.3 — Metrics That Matter

Concept target	Measurement, proxy choice, incentives, how a metric changes a ranking.
Source activity	Metric Match and Agent's Playbook; choose which metric best supports a valuation argument.
V5 classroom form	Metric Builder. Students choose or weight two simple measures, watch the ranking reorder, and defend the metric against a stated team goal.
Class-generated data	Metric selected, weights, rank changes, player chosen, goal fit, disagreement across teams.

Second run / comparison	Introduce a player who games the metric or change the team goal. The point is not one perfect metric; it is a defensible fit and boundary.

Module 4 — Draft Day Basics

101.4.1 — Why the Draft Isn't a Ranking

Concept target	Uncertainty, outcome ranges, process versus result.
Source activity	Current flagship repeats the same strategy and produces scores from 40 to 93 while reporting process and outcome separately.
V5 classroom form	Draft Distribution Lab. Students choose a process, the class runs many parallel drafts, and the display plots the outcome distribution rather than crowning one lucky winner.
Class-generated data	Strategy, process score, outcome score, distribution, upset rate, confidence before and after.
Second run / comparison	Rerun the same strategy with a fresh random seed. Compare what changed and what did not.

101.4.2 — Floor, Ceiling & Prospect Pyramid

Concept target	Risk profile, volatility, projectability, portfolio logic.
Source activity	Source supplies five prospects with floor, ceiling, and volatility descriptions.
V5 classroom form	Prospect Portfolio. Teams allocate three picks across safe, balanced, and high-variance profiles for a stated roster goal, then simulate many futures.
Class-generated data	Portfolio profile, expected contribution, downside, upside, volatility, goal fit, chosen rationale.
Second run / comparison	Change the team from contender to rebuilding. Students must revise the same prospect board rather than receive a new answer key.

101.4.3 — Fit vs. Best Player Available

Concept target	Context-dependent strategy, complementarities, roster timeline.
Source activity	Current flagship has no dominant heuristic: rebuilders and contenders reward different choices.
V5 classroom form	Draft Room Fit vs. BPA. Preserve the decision core; add a shared class comparison by team context and a short role-based front-office discussion.
Class-generated data	Choice by context, fit score, talent score, timeline, strategy pattern, revision after reveal.
Second run / comparison	Switch the roster context while keeping the board constant. A rule that survives every context should be treated skeptically.

11. Track 201 lesson-by-lesson V5 map

Track 201 should become denser and more strategic without turning into an arithmetic exam. Students can manage more variables, private information, uncertainty, model disagreement, multi-period consequences, and second-order effects. The interface calculates; students reason.

Module 1 — Cap Crash

201.1.1 — How Teams Bend the Cap

Concept target	Exceptions, contract engineering, rule interpretation, legal constraint.
Source activity	Cap Crash challenge uses Bird Rights, exceptions, and roster legality; current open sandbox rewards value over star stacking.
V5 classroom form	Cap Crash Control Room. Teams inherit different rosters and owner mandates, then engineer legal solutions using a rule engine that explains which constraint each move touches.
Class-generated data	Moves, exception use, legal status, roster depth, expected wins, future obligations, paths attempted.
Second run / comparison	A new injury or retention need arrives after lock-in. Compare teams that preserved versus spent flexibility.

201.1.2 — Luxury Tax in Action

Concept target	Marginal tax, expected benefit, owner risk, decision under uncertainty.
Source activity	Owner Risk Meter asks students to hit wins, spend, and performance targets.
V5 classroom form	Tax Curve. Students choose upgrades under private playoff probabilities and an owner mandate; the display separates payroll, marginal tax, total cost, and expected competitive value.
Class-generated data	Scenario, probability assumption, chosen moves, tax curve position, EV, owner tolerance, rejected alternative.
Second run / comparison	Change only the probability or owner objective. Students see that the same roster move can become rational or irrational.

201.1.3 — Second Apron

Concept target	Multi-period constraint, path dependence, option value, restrictions.
Source activity	Source lesson explains the second apron and the loss of transaction tools; activity is weak or absent.

V5 classroom form	Two-Year Cap Planner. Teams choose a year-one roster, then face year-two renewal and trade scenarios. Restrictions are modeled as unavailable actions, not a text warning.
Class-generated data	Year-one performance, future payroll, actions lost, replacement quality, flexibility index, final explanation.
Second run / comparison	Compare two teams with similar year-one performance but different option sets. Debrief flexibility as a measurable asset.

Module 2 — Money in Motion

201.2.1 — The League as a Business

Concept target	Joint venture, media rights, revenue sharing, bargaining, collective governance.
Source activity	Analyze a media deal and its cap/payroll impact.
V5 classroom form	Media Rights Deal Room. Student roles represent large-market teams, small-market teams, players, league office, and broadcaster; a deal must satisfy a stated coalition rule.
Class-generated data	Offers, coalition membership, revenue split, cap impact, market distribution, failed deals, concessions.
Second run / comparison	Release new streaming information or change the sharing rule. The second run tests who held leverage and why.

201.2.2 — Player Economics

Concept target	Guarantees, incentives, duration, expected value, bargaining power, risk allocation.
Source activity	Contract Builder / Agent Simulator compares safe and upside offers; MLB Agent Simulator makes counterfactual EV explicit.
V5 classroom form	Contract Table. Agent and team receive different private information about health risk, market demand, and owner constraints; they negotiate money, guarantees, incentives, and term.
Class-generated data	Offers, counteroffers, guarantees, incentives, EV by party, walk-away, information revealed, explanation.
Second run / comparison	Reveal or withhold a medical signal, or improve the player’s outside option. Compare value created and risk shifted.

201.2.3 — The Ownership Equation

Concept target	Long-term asset value, cash flow, investment, risk, time horizon.
Source activity	The current source explicitly has no simulation and presents ownership as a mindset.

V5 classroom form	Franchise Value Lab. Teams allocate scarce capital among roster, venue, media, community, and operations across several periods; uncertain returns expose the difference between cash flow and asset appreciation.
Class-generated data	Capital allocation, cash flow, franchise value components, risk exposure, fan trust, opportunity cost.
Second run / comparison	A media or arena shock changes the value of earlier investments. No universal 'owner mindset' conclusion is allowed.

Module 3 — Analytics in Action

201.3.1 — Data to Decisions

Concept target	KPI selection, ROI, tradeoffs, decision dashboards.
Source activity	Front Office Dashboard Builder; current flagship punishes famous-name bias and adds owner mandates and a trade-deadline adaptation.
V5 classroom form	Front Office Dashboard. Preserve the strongest existing core; separate evidence selection from player selection and make the owner mandate visible as a constraint.
Class-generated data	KPIs chosen, player/roster decision, projected ROI, actual season, deadline move, strategy revision.
Second run / comparison	Swap the mandate or hide one KPI. Debrief which measure changed the decision and whether it deserved that power.

201.3.2 — Modeling the Market

Concept target	Expected value, forecast uncertainty, aging/risk, model disagreement.
Source activity	Projection sliders and model-building activities; Decision Room already presents two dashboards with named blind spots and an injury shock.
V5 classroom form	Model Room. Teams receive two credible models that disagree, choose what to trust, make a signing, and then respond when a new signal breaks the favored story.
Class-generated data	Model chosen, variables weighted, confidence, decision, forecast range, error, revision, explanation of blind spot.
Second run / comparison	Run the same case with different missing data. The goal is calibrated confidence, not beating the forecast.

201.3.3 — Competitive Edges

Concept target	Capability investment, data as capital, complementarities, organizational execution.
Source activity	Organizational Draft / Front Office Draft: build an analytics team under a budget.

V5 classroom form	Analytics Organization Draft. Teams allocate budget among data, scouts, infrastructure, communication, and implementation; returns depend on complements rather than one superstar hire.
Class-generated data	Budget mix, capability gaps, adoption, forecast quality, decision speed, value realized over time.
Second run / comparison	Change the organization's starting weakness or staff culture. A good team in one context should not dominate every context.

Module 4 — Draft Economics

201.4.1 — True Value of a Draft Pick

Concept target	Expected value curve, probability, nonlinearity, market price versus underlying value.
Source activity	Pick Value Simulator compares fake charts with real data; GM Efficiency Frontier models marginal value honestly.
V5 classroom form	Draft Asset Market. Teams receive pick distributions and roster windows, then bid or trade for picks. The class market price is compared with expected value after trading closes.
Class-generated data	Bids, asks, prices, EV, roster window, surplus paid/received, forecast confidence.
Second run / comparison	Change the prospect class distribution or information precision. Compare whether prices adapt or herd.

201.4.2 — Rookie Contracts & Surplus Value

Concept target	Surplus value, rookie scale, cap allocation, portfolio risk.
Source activity	Surplus Value Engine; current NBA Surplus Value Championship combines surplus and synergy.
V5 classroom form	Surplus Portfolio. Teams build a roster from rookies and veterans under cap and role constraints; cheap talent is valuable but uncertain and complementary fit matters.
Class-generated data	Contract cost, expected contribution, surplus, synergy, volatility, depth, concentration risk.
Second run / comparison	Introduce an injury or development miss. Compare diversified and concentrated surplus strategies.

201.4.3 — Trade Up / Trade Down

Concept target	Expected value, information asymmetry, roster window, opportunity cost.

Source activity	Timed trade decisions; current Trade Deadline War Room 201 makes the identical trade right or wrong by team and tax context.
V5 classroom form	Draft War Room. Teams receive different windows and private beliefs, then negotiate trade-up, trade-down, or hold decisions under a closing market.
Class-generated data	Offers, accepted trades, EV delta, tax/cap effect, window fit, alternatives forgone, final reasoning.
Second run / comparison	Swap only the roster window. The debrief shows why the same trade cannot be scored without context.

12. Lesson architecture library

A single fixed sequence would make the product easier to template and worse to teach. V5 uses six named lesson forms. Every form still protects three moments: a student commits before the answer is known; the system produces interpretable feedback; the lesson closes with explanation.

Lesson form	Best for	Typical flow
Predict → system → reveal → theory	Emergent patterns and misconceptions	Hook; prediction; short rules; run; class graph; explain; exit.
Teach a little → run → teach → rerun	A rule students can feel before formalizing	Mini-model; round 1; reveal; theory; one rule change; round 2; compare.
Role briefing → negotiation → debrief	Private information, bargaining, coalition	Public goal; private cards; offers; agreement/failure; reveal hidden structure; counterfactual.
Build → shock → adapt	Cap, portfolio, operations, path dependence	Construct; lock; shock; revise; compare resilient versus brittle choices.
Competing models → decision → new signal	Analytics, uncertainty, evidence	Inspect models; choose; commit; receive signal; revise; calibrate confidence.
Institution design → simulate → defend	Policy, revenue sharing, caps	Choose objective; set rule; run; inspect distribution; defend and critique.

Default 55-minute Ramaz rhythm

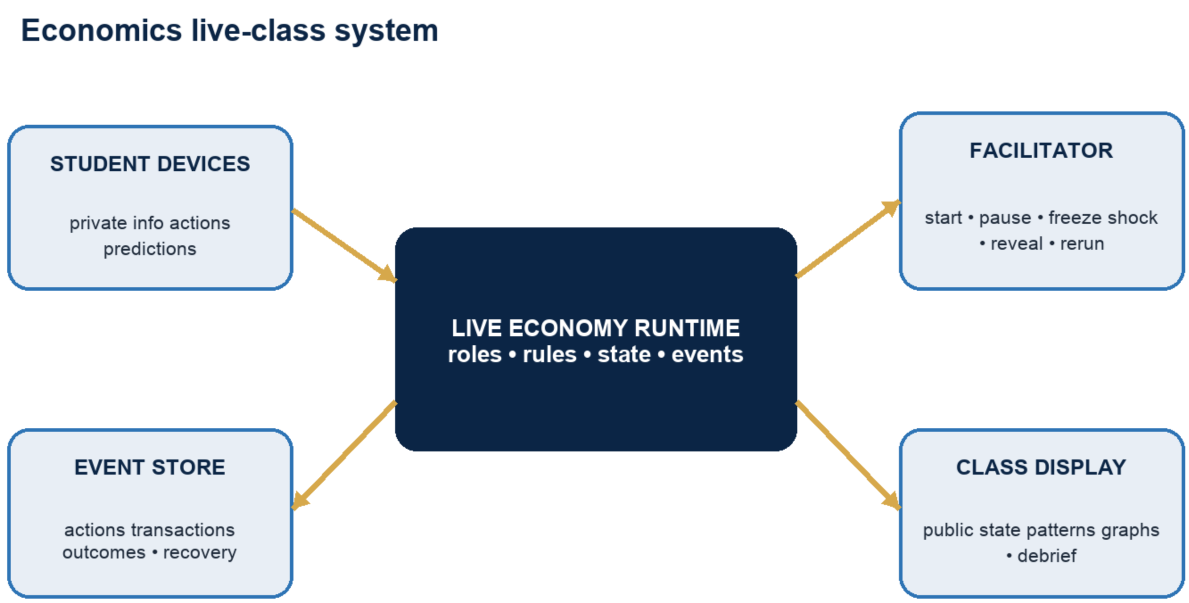
Minute	Room state	Job
0–5	Hook + prediction	Create the problem and capture an initial belief.
5–12	Compact teaching	Give only the model and vocabulary needed to decide.
12–15	Join + role	Code, name/alias, role, one-screen rules, readiness.
15–27	Round 1	Students act; Brayden monitors patterns and room energy.
27–34	Reveal	Freeze; show the class data; ask what happened before explaining why.
34–40	Theory	Name the mechanism, limits, and competing interpretation.

Minute	Room state	Job
40–49	Change / Round 2	Change one economically meaningful condition and rerun or compare.
49–55	Debrief + exit	Compare, explain, and record one claim or unresolved question.

NOT A STOPWATCH

The rhythm is a default, not a template prison. A negotiation may need one long run; a draft-distribution lab may need several short runs; a direct lesson may use a supporting activity instead of a simulation.

13. Ramaz classroom operating model



The live runtime is the connective tissue among student action, facilitator control, class display, and recoverable event data.

Room assumptions

- One Brayden facilitator, a shared display, and one device per student or team.
- A normal class lasts 50–60 minutes; join friction must consume seconds, not a lesson segment.
- Accounts are off by default. Use a short session code, safe display name or alias, and an assigned role.
- The network, a device, or a student will fail. Recovery and role reassignment are part of the product, not an operations footnote.

- Controlled productive chaos is allowed. The facilitator must be able to freeze the economy and reclaim attention immediately.

Student join flow

1. Open the stable Economics join URL or scan the QR code.
2. Enter the four-to-six-character class code.
3. Enter a first name, initials, or teacher-assigned alias according to the session privacy setting.
4. Receive a role, private information, and one sentence explaining the goal.
5. Complete one readiness check and wait in the ready room. Target median join time: under 60 seconds.

Grouping rules

Structure	Use when	Avoid when
Individual	The concept is forecasting, measurement, or personal strategy and peers do not change the state.	A shared market or negotiation needs counterpart behavior.
Pairs	The reasoning benefits from talk and one device; useful in Track 101 and negotiation teams.	Both students need private, opposing information.
Teams	Roles and organizational tradeoffs matter; teams can divide analyst/scout/owner work.	A dominant student can hide everyone else's reasoning.
Whole-class network	Trades, prices, public goods, coalitions, or shared resources make interdependence the concept.	The interaction is only simultaneous single-player.

Controlled productive chaos protocol

Before	State the noise rule, movement rule, time signal, freeze cue, and what remains private.
During	Public display shows only the information students need; private values, costs, or offers stay on devices.
Freeze	One action stops new submissions, preserves state, mutes timers, and displays a neutral attention screen.
Recover	Brayden can reassign a role, activate a bot, reopen a student, or invalidate a malformed round without rebuilding the class.
After	The display changes from play state to evidence state. Students stop acting before class metrics are interpreted.

14. Facilitator control room

The facilitator should feel like a game master with a small set of meaningful levers, not a system administrator watching telemetry. The control room has three layers: always-visible run controls, context controls supplied by the simulation contract, and diagnostics hidden until needed.

Layer	Controls	Design rule
Always visible	Start; pause; freeze; advance/end round; reveal; rerun; show class data.	Large, stable placement; actions confirm the resulting room state.
Simulation controls	Open/close market; release information; apply a named shock; change a policy; set timer; choose comparison.	Only pre-authored controls with a truthful economic interpretation.
Recovery	Reassign role; add bot; remove duplicate; reconnect device; reopen submission; mark invalid round; restore snapshot.	Hidden by default but reachable in one step.
Facilitation intelligence	Readiness, stuck students, action distribution, no-trade warning, timing suggestion.	Describe observable state; do not prescribe the economic conclusion.

Control semantics

PAUSE	Stops the timer and new state transitions but leaves the current student screen visible.
FREEZE	Stops action and switches every student to a neutral attention state; state is preserved.
REVEAL	Publishes a defined subset of hidden information or class metrics. It never exposes a student's private value by name.
SHOCK	Applies a versioned, economically valid scenario change with an instructor preview of affected variables.
RERUN	Creates a linked comparison round that pins the same participants and starting state unless the spec declares otherwise.
END	Closes actions, writes a final snapshot, and opens the debrief recipe. It is reversible only through an explicit restore.

Anti-cockpit rules

- No more than seven primary controls during live play.
- No raw event stream, database vocabulary, or developer errors in the teaching surface.
- No shock button without an explanation of what changes and a preview of the data reset policy.

- No automatic AI interpretation shown as fact. Suggestions must cite the observable class pattern that triggered them.
- No control that changes a student's choice after commitment without a visible audit record.

15. Live class display

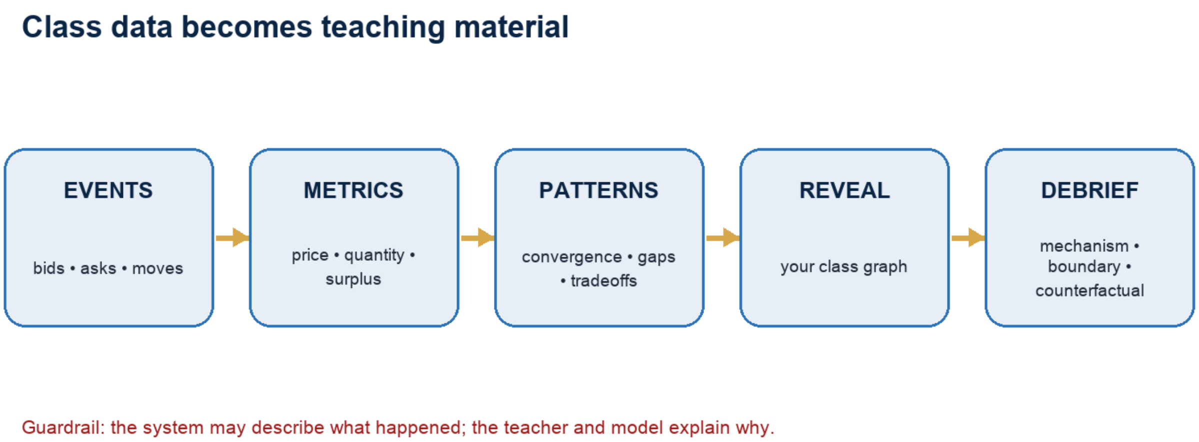
The display is not a mirrored student screen or an always-on leaderboard. It is a staged public instrument. Each state has one room-level job.

Display state	Shows	Protects
Join	Short URL/QR, code, count joined, help cue.	No student names unless the teacher enables them.
Ready room	Roles filled, teams ready, one-sentence goal, start signal.	No private values, costs, model cards, or strategies.
Prediction	Anonymous distribution of initial beliefs.	No correct-answer cue.
Live system	Public market/resource state, round clock, aggregate activity, allowed announcements.	No personally identifying performance ranking.
Freeze/event	Clear attention screen and the exact rule or shock change.	No moving data behind the explanation.
Class data	The metrics and graph defined by the debrief recipe.	No causal label unless the comparison supports it.
Compare	Round 1 versus Round 2 or group A versus group B with starting differences visible.	No false comparability.
Debrief	Pattern, model, boundary, student question, next decision.	No score theater.

Privacy on the big screen

Default to anonymous aggregates. A student may be spotlighted only through an opt-in safe alias and only when the learning value exceeds the social cost. Private willingness to pay, cost, contract constraint, medical signal, or bargaining position must never be projected during play.

16. Class-generated data and debrief system



The debrief engine transforms events into a small teaching view. It does not manufacture causal certainty.

The event-to-debrief chain

1. Capture versioned student, system, facilitator, and technical events.
2. Calculate only metrics declared in the simulation's class-data contract.
3. Detect notable patterns against transparent thresholds, not opaque 'insight' scores.
4. Render one to three graphs chosen for the concept and class age.
5. Show what happened before showing the model that may explain it.
6. Offer comparison and counterfactual prompts only when starting conditions make them interpretable.
7. Close with a student explanation or unresolved question; preserve a compact session receipt.

Minimum class data schema

Record	Fields	Use
Session	class/session ID; experience and version; motif; start/end; device count; privacy mode.	Reproduce and interpret the run.
Participant/role	ephemeral participant ID; team; role; constraints; private-info version.	Understand available choices without storing unnecessary identity.
Prediction	prompt version; response; confidence; time.	Compare belief before and after.
Action	action type; parameters; state before/after; validity; time.	Reconstruct choices and strategies.
Transaction	counterpart IDs/teams; offer; acceptance; price/quantity/terms.	Market and bargaining metrics.

Record	Fields	Use
Snapshot	round; public state; hidden-state hash; rule/policy version.	Recovery and comparison.
Facilitator	pause/freeze/reveal/shock/reassign/restore; reason when destructive.	Interpret interventions and debug the lesson.
Outcome	declared metrics; uncertainty; comparison set; invalidity flag.	Power the class display and debrief.
Explanation	prompt; selected claim or short text; revision link.	Capture reasoning without inventing a grade.
Technical	disconnect; retry; latency; recovery; client version.	Separate product failure from student behavior.

Debrief recipe contract

Before	One prediction and the misconception or tradeoff it is intended to expose.
During	The events and metrics required; what may remain hidden; invalid-run criteria.
Reveal	One headline pattern, up to three metrics, up to two graphs, and a plain-language observation.
Theory	Mechanism, causal assumptions, model boundary, and connection to the lesson objective.
Counterfactual	The single rule/input changed, what remains constant, and what comparison is defensible.
Questions	Notice; explain; challenge; transfer. Suggested wording, never a script Brayden must read.
Receipt	Session version, participation, outcome summary, notable technical issue, and one student explanation sample.

Data truth rules

- Show sample size and invalid or missing actions.
- Separate observed result, model interpretation, and instructor hypothesis visually.
- Do not compare groups that had different rules without naming the difference.
- Do not present one stochastic outcome as proof that a strategy is better.
- Do not calculate precision the underlying model cannot support.
- Do not retain student names merely because the event system can.

17. Economics mechanics taxonomy

The current library is rich in roster allocation, drafting, and front-office scenarios and comparatively thin in market formation, production, public goods, labor, competition, and externalities. The taxonomy below becomes both a design vocabulary and a portfolio-coverage test.

Mechanic family	Student verbs	Makes visible	Best grouping
Allocation under constraint	allocate, build, choose	Scarcity, opportunity cost, complementarity, flexibility	Individual / team
Continuous market	bid, ask, buy, sell	Price discovery, gains from trade, convergence, shortage	Whole class
Auction	value, bid, wait	Willingness to pay, winner's curse, information	Whole class / teams
Bargaining & coalition	offer, counter, reveal, ally	Leverage, BATNA, surplus division, collective rule	Pairs / teams
Trade & exchange	specialize, trade, hold	Comparative advantage, transaction cost, mutual gain	Whole class
Production system	hire, produce, invest, schedule	Productivity, fixed/variable cost, bottleneck	Teams
Competition & entry	price, enter, differentiate	Market power, strategic response, profit erosion	Teams
Public goods & commons	contribute, extract, enforce	Free riding, depletion, institution design	Whole class
Information & signaling	inspect, disclose, infer, screen	Adverse selection, moral hazard, signal quality	Pairs / network
Forecasting & model risk	weight, forecast, calibrate, revise	Uncertainty, bias, disagreement, error	Individual / team
Shock & adaptation	commit, respond, reallocate	Resilience, path dependence, option value	Any
Portfolio & intertemporal	invest, diversify, defer	Risk-return, compounding, horizon, concentration	Individual / team
Coordination & network	choose standard, coordinate, switch	Equilibrium, network effect, lock-in	Whole class
Institution design	vote, regulate, tax, subsidize	Objective tradeoffs, incentives, unintended effects	Teams / whole class

Current coverage judgment

PORTFOLIO GAP

BOW can teach the current sports-business curriculum well, but it does not yet demonstrate the breadth of an Economics simulation system. The first future-curriculum prototype should be a live double-auction market because it stress-tests real-time roles, private values, transactions, class graphs, price convergence, tax/shock reruns, and facilitator orchestration in one vertical slice.

18. Motif and World architecture

Sports remains BOW Economics' flagship motif because it gives the current curriculum authentic rules, assets, contracts, uncertainty, and social energy. It should not become a permanent limit or a decorative skin applied to mechanisms it distorts.

Level	Definition	Comparability rule	Example
Flavor	New nouns, art, audio, or story with the same roles, variables, and decision contract.	Same version family only if content and behavior are truly unchanged.	Basketball tickets versus concert tickets in an otherwise identical posted-price case.
Case variant	New data or scenario changes the decision while preserving the evidence target.	Comparable only through a validated comparison-set version.	Contender versus rebuilding team in Fit vs BPA.
Motif	A sustained thematic expression with distinctive roles and interface language.	Mechanics may be shared; experience meaning is not assumed identical.	Sports front office, creator economy, city systems.
World	A coherent institution and narrative space with multiple experiences, assets, and progression.	No cross-World comparability by default.	A future Creator Economy World with contracts, platforms, audiences, and revenue systems.

Mechanic-first selection

1. Name the economic concept and the misconception or relationship students must experience.
2. Choose the mechanism that makes the relationship manipulable, consequential, and observable.
3. Choose the motif whose real institutions make that mechanism easiest to understand.
4. Write the model boundary and remove story elements that imply false economics.
5. Only then design art direction, terminology, narrative, and content variants.

Motif roadmap

Now	Sports as flagship; neutral market/city contexts when sports creates unnecessary rules or licensing/staleness risk.
Next candidate	Creator economy, because it supports platforms, contracts, pricing, audience/network effects, sponsorships, and risk. Validate one concept before declaring a World.

Other candidates	Food/retail for production and pricing; cities for public goods and policy; music/entertainment for contracts and demand. These are hypotheses, not a build list.
Trigger	A second motif is productized only after one shared mechanic delivers equal or better learning, student pull, and debrief quality without confusing comparability.

World contract

A World declares its concepts, roles, institutions, recurring resources, terminology, aesthetic system, accessibility notes, content/version owner, and forbidden claims. It may implement shared mechanic interfaces, but it owns its narrative meaning. A World is not a folder of reskins.

19. Simulation Library refoundation

The library should stop presenting repository discovery as product readiness. Its job is to help an instructor find the right experience, understand whether it will run, and know exactly how to facilitate and debrief it.

Recommended public hierarchy

Shelf	Promotion rule	Audience
Ramaz-ready	Live-run evidence; current deployment; facilitator recipe; recovery path; debrief data; no P0 blockers.	Brayden now; later trained instructors.
Core curriculum	Mapped to a Track/module/lesson with a clear learning job; may be in development.	Curriculum and product team.
Supporting activities	Useful case, quiz, reflection, or short individual practice; not represented as a simulation flagship.	Instructors selecting a lighter lesson form.
R&D bench	Promising mechanic, concept, or World not yet ready for teaching.	Design and engineering.
Archive	Superseded, unreachable, unmapped, or historically useful; preserved with provenance.	Internal reference.

Simulation detail page

- Learning job: concept, objective, misconception, model boundary, Track/module/lesson.
- Class job: duration, grouping, devices, first decision time, instructor actions, room-energy level.
- Run job: verified launch, version, health, fallback, data policy, accessibility, current owner.
- System job: agents, incentives, constraints, information, actions, feedback, dynamics, player verbs.

- Debrief job: required events, metrics, graphs, questions, comparison rule, invalid-run criteria.
- Governance job: curation decision, rationale, live receipts, known blockers, change log, promotion gate.

Registry rules

- One canonical record per experience; immutable versions for any session used in teaching.
- A deployment is not 'healthy' because the URL returns a page. Health includes the full critical path.
- A concept tag does not prove instructional alignment. The lesson owner must declare the experience's actual job.
- Every public launch has an owner, last-verified date, fallback, and retention policy.
- Automated checks can detect reachability, build failures, stale versions, and missing metadata; humans certify economics, facilitation, and student pull.

20. Simulation audit: portfolio decision

All 76 discoverable records were classified. The result is deliberately severe because the goal is not to preserve count. KEEP means preserve the current core while validating it live; it does not mean the experience is finished.

Decision	Count	Meaning
KEEP	9	Strong current core; validate live and repair only evidence-backed issues.
REFINE	14	Good interaction or model; targeted product, data, balance, or facilitation work.
REFOUND	10	Concept or premise is valuable; current system cannot carry V5.
MERGE	16	Combine duplicated concepts, builds, or mechanic lines into a stronger canonical experience.
SUPPORTING ACTIVITY	4	Useful teaching asset that should not be marketed as a flagship simulation.
ARCHIVE	21	Preserve provenance; remove from the active teaching surface.
KILL	2	No runnable experience or salvage value proportional to maintenance.

Provisional KEEP set

Experience	Why the core survives	Immediate use
Cap Crash	Open sandbox; value beats star stacking; legal roster rule blocks the obvious exploit.	201 M1 foundation.

Experience	Why the core survives	Immediate use
Draft Room Fit vs BPA	No universal heuristic survives different team contexts.	101 M4 L3.
Front Office Dashboard	Real traps, changing mandate, season, and deadline adaptation.	201 M3 L1.
Mega City Tycoon	Binding civic budget and reflection tied to skipped projects.	Future policy / allocation shelf.
MLB Agent Simulator	Expected value and counterfactuals are visible before commitment.	201 M2 L2 core to refound socially.
MLB Money Maker	Multi-stakeholder bargaining cannot be solved by one lever.	201 M2 extension / negotiation kernel.
Small Markets, Big Money	Stakeholder deltas before commitment and genuine branching.	101 M2 L2.
Trade Deadline War Room 201	Same trade changes meaning with tax and roster context; strong mobile build.	201 M4 L3.
Why the Draft Isn't a Ranking	Repeated outcomes prove process and result are different.	101 M4 L1.

What the audit says about the current portfolio

- The strongest assets are decision systems, not the most polished course wrappers.
- Several excellent concepts are broken, unreachable, duplicated, or trapped in spreadsheet/server dependencies.
- Many experiences are individual and self-guided; the next-generation live class layer does not yet exist.
- Quiz-first experiences can remain useful diagnostics or supporting activities but should stop carrying simulation branding.
- Track 301 contains some of the strongest model-risk and portfolio work; treat it as an R&D shelf, not an immediate Ramaz scope expansion.
- Financial Literacy is too thin in this repository to infer its next product directly; use the separate Decision Challenges architecture.

VALIDATION LIMITATION

Direct new browser playtesting from this workspace was blocked by an administrator-enforced browser policy. V5 therefore uses the Simulation Library's recorded end-to-end playtests and source metadata and does not claim fresh direct verification. Browser revalidation remains a promotion gate.

21. Flagship Economics portfolio

The next-generation portfolio should be organized around signature economic experiences, not one bespoke build per slide. Eight flagships cover the current tracks; one live-market prototype expands the mechanic system and proves the platform.

Flagship	Curriculum job	Mechanic	Starting assets
Cap Builders	101 M1: rules, tax, flexibility	Allocation + shock/adapt	Front Office Build the Roster; Cap Crash; Pregame concept
League Balance Lab	101 M2: revenue, sharing, cap purpose	Institution design + simulation	Small Markets, Big Money; League in a Box
Value & Signal Room	101 M3: efficiency, evidence, metrics	Draft + information + metric builder	Moneyball Draft; Stats vs Scouts; Analytics Lab / Stat Inventor
Draft Uncertainty Lab	101 M4: distributions, risk, fit	Stochastic selection + portfolio	Why Draft Isn't Ranking; Fit vs BPA; Process vs Results
Cap Crash Control Room	201 M1: engineering and multi-period constraint	Open allocation + rules + shock	Cap Crash; Luxury Tax; Curve Room
League & Contract Deal Room	201 M2: revenue and contract bargaining	Coalition + bilateral negotiation	MLB Money Maker; MLB Agent; Boss Summit premise
Model Room	201 M3: KPIs, forecasts, capability	Competing models + new signal	Front Office Dashboard; Decision Room; analytics pair
Draft Asset Market	201 M4: EV, surplus, trade window	Auction/trade + portfolio	Trade Deadline 201; GM Trade; Surplus Value
Your Market	Future Economics / Simulation Day	Live double auction + policy rerun	New vertical slice; MobLab/EconPort pattern benchmark

Build versus use

Tier	Use
Use now with facilitation	The nine KEEP cores plus selected REFINE activities, with clear health and fallback notes.
Prototype now	Your Market, Cap Builders, League Balance Lab: together they test real-time trading, team allocation, institution changes, display, control, data, and rerun.
Refound after first receipts	Draft Uncertainty, League & Contract Deal Room, Model Room, Draft Asset Market.
Do not build yet	A universal motif marketplace, Track 301 product, Economics Decision Challenge line, or migration layer for all legacy repos.

22. Prototype priorities

Prototype uncertainty, not screens already understood. Each prototype must include the student action, facilitator control, display reveal, event capture, and debrief. A player-only demo cannot answer the Ramaz product question.

Prototype question	Competing versions	Decision evidence
How should a live market feel?	Continuous bid/ask book vs. periodic clearing; visible order book vs. private offers.	First decision time, trade volume, convergence, confusion, debrief clarity.
How should Track 101 build a roster?	Card-drag roster vs. constrained choice rounds; individual vs. pair.	Rule comprehension, arithmetic load, option visibility, room pace.
How should a rule shock land?	Instant state change vs. freeze/rebrief/reopen; one shared shock vs. differentiated shocks.	Prediction quality, fairness, recovery, comparison interpretability.
What belongs on the class display?	Aggregate-only vs. public market state; chart-first vs. story-first reveal.	Attention, strategic leakage, ability to notice pattern before explanation.
How should private information work?	Role card, incremental reveal, or hidden data tab.	Leaks, accessibility, strategic use, teacher recovery.
How should disconnected players recover?	Rejoin by token, facilitator reassignment, bot takeover.	State continuity, time lost, data validity.
Can motifs remain comparable?	Sports tickets vs. concert tickets using one market contract.	Equivalent comprehension and outcome distributions; language does not change strategy unfairly.

Prototype kill tests

- A student can succeed by clicking a highlighted or consistently positioned option.
- The first consequential action arrives after five minutes of interface explanation.
- A live multiplayer build is functionally simultaneous single-player.
- The teacher sees only a score or must inspect raw student screens to debrief.
- Changing the rule changes the story copy but not the economic state transition.
- Luck determines the lesson's claimed conclusion and the display hides the distribution.
- The visual skin is premium but the student verbs remain read → click → read.

PART III

Build, prove, and transfer

Visual direction, slide integration, evidence gates, roadmap, and the adversarial review that the implementation must survive.

23. Visual and interaction direction

The product should feel like a premium interactive system built for a room, not a worksheet, generic dark dashboard, or sports-game imitation. Information hierarchy and consequence visibility matter more than decoration.

Surface	Visual job	Direction
Student	Make role, objective, constraint, available action, and consequence immediately legible.	Tactical card/panel composition; one primary action; private information clearly marked; consequences animate briefly and remain inspectable.
Facilitator	Keep room state and next action calm under pressure.	Light or quiet dark-neutral utility surface; large stable controls; warnings tied to recovery actions; no ornamental motion.
Class display	Hold shared attention and reveal the class economy.	Broadcast-scale typography; cinematic state transitions; one story per screen; graphs large enough to read from the back.
Library	Communicate readiness and teaching job before visual excitement.	Clear curation badges, verified date, lesson map, launch/fallback, screenshots, short debrief preview.

Track differentiation

Track 101	Broadcast clarity: larger type, fewer simultaneous variables, concrete icons, explicit state changes, warm energy, pair-friendly interaction.
Track 201	Front-office depth: denser comparison, ranges and uncertainty, private information, multi-period state, model disagreement—without spreadsheet clutter.
Shared	Same feedback grammar, accessibility, session states, control semantics, and visual distinction among public, private, estimated, and final information.

Feedback grammar

Feedback type	Treatment
Action accepted	Immediate state confirmation and a visible change in the variable affected.
Action blocked	Name the rule or constraint, show the missing condition, preserve the student's intent.
Uncertain outcome	Show range/probability before the reveal and do not convert it into false certainty afterward.
Counterpart response	Make the response attributable to incentives or information, not arbitrary personality text.
Collective consequence	Reveal the system-level pattern after individual actions lock; distinguish the class result from the model expectation.
Revision	Keep the first choice visible and connect the revision to the new information or rule.

Accessibility and performance

- Keyboard-complete critical path; no drag-only decision.
- Text alternatives for every graph and non-color indicators for state and role.
- Reduced-motion mode; motion communicates state, never hides content or creates pressure without a learning reason.
- Chromebook-first performance and responsive layouts; public display tested at classroom projection distance.
- Timers are visible, pausable, and never the sole source of difficulty.
- Private information has screen-reader labels that do not leak through the class display.

24. Future slide redesign specification

DO NOT REDESIGN YET

Slides should be rebuilt only after each lesson's new role, mechanic, data reveal, counterfactual, and debrief are accepted. The current decks are intentionally not inputs to V5's lesson architecture.

Slide roles

Role	Purpose	Live integration
HOOK	Create a concrete tension or role.	May become the session start screen.
TEACH	Introduce one model, rule, or vocabulary set.	Static or lightly interactive; no giant paragraph.
PREDICT	Capture a belief before play.	Student responses can populate the display.

Role	Purpose	Live integration
JOIN	Move the room onto devices.	Stable URL/QR/code and readiness count.
RULES	Show goal, action, constraint, and privacy boundary.	Mirrors the experience version.
LIVE STATE	Public system status during play.	Rendered from runtime, not manually updated slides.
EVENT	Explain the shock or rule change.	Triggered from facilitator control.
DATA REVEAL	Show the class outcome.	Generated from the debrief recipe.
MODEL	Explain the mechanism and boundary.	Can annotate the class graph rather than replace it.
COMPARE	Round 1 vs. Round 2 or model expectation vs. observed.	Starting differences must be visible.
DEBRIEF	Notice, explain, challenge, transfer.	Captures one closing explanation or question.

Slide workstream inputs

1. Accepted lesson blueprint and concept/model boundary.
2. Student, facilitator, and display state maps.
3. Final simulation rules, data contract, and debrief recipe.
4. Current slide deck for content salvage and provenance—not structural authority.
5. A visual storyboard for live transitions and failure/fallback states.
6. A rehearsal script with timing, expected misconceptions, freeze points, and optional branches.

Slide quality rule

Brayden carries the voice. Slides carry the visual model, question, role, rule, live state, or evidence that the room needs at that moment. Some explanatory text is appropriate; paragraphs that compete with live facilitation are not.

25. Implementation architecture for Economics

Build one authoritative live-session product with three views and composable economy kernels. The kernel owns rules and state; the views own presentation. Legacy simulations remain linked or wrapped until their mechanic is deliberately migrated.

Layer	Responsibility	Non-negotiable
Player app	Join, role/private info, action, feedback, revision, reconnect.	Fast first action; keyboard; no hidden dependency on teacher account state.
Facilitator app	Session setup, grouping, run control, valid shocks, recovery, debrief launch.	One instructor can operate while teaching.

Layer	Responsibility	Non-negotiable
Display app	Public state, prediction, event, metrics, graphs, comparison, debrief.	Never leaks private or personally identifying data.
Session engine	Authoritative state, timing, roles, validation, snapshots, reconnect, bots.	Deterministic replay where randomness is seeded and recorded.
Economy kernels	Simulation-specific agents, actions, transitions, metrics, model boundaries.	Pure, testable rules separate from UI.
Event/debrief	Versioned events, declared metrics, recipes, receipts, invalid-run flags.	Description and interpretation remain distinct.
Registry adapter	Read canonical simulation, lesson, version, health, run resource, and promotion data.	The registry never becomes the live runtime.

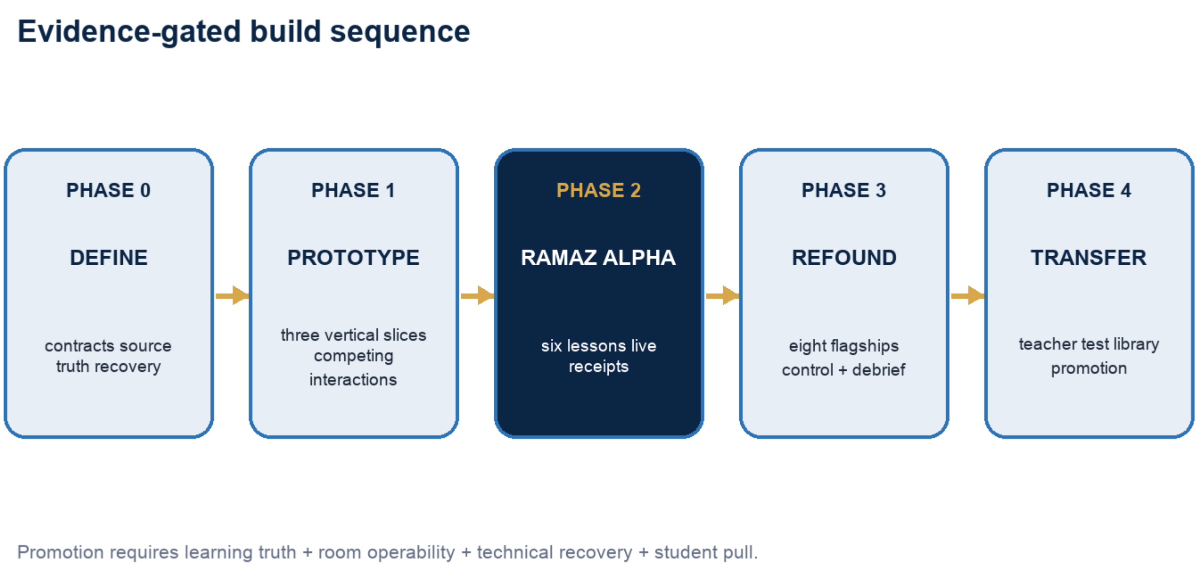
Reliability requirements

- Authoritative server state; clients cannot award themselves resources or submit impossible actions.
- Seeded randomness recorded with every outcome; a stochastic result can be reproduced.
- Snapshot before start, before every shock, and at round close; restore available to the facilitator.
- Reconnect restores role and state without a new identity; facilitator can replace a missing role with a bot.
- Graceful degraded mode for display loss and a printable/offline fallback for each Ramaz-ready lesson.
- Load test above the expected class size with weak-network simulation and device throttling.
- No sensitive student data in logs; retention is explicit and session identity is ephemeral by default.

Do not decide the vendor too early

The prototype must compare real-time hosting options against session recovery, classroom latency, operational simplicity, and cost. The product contract should remain portable: authoritative rooms, events, snapshots, and reconnect are requirements; a specific database or messaging vendor is an implementation choice.

26. Evidence-gated roadmap



The roadmap promotes vertical slices only when learning and room evidence are strong enough—not when code is merely complete.

Phase	Deliverable	Exit gate
0 — Define	Curriculum registry; five contracts; three flagship specs; privacy/accessibility policy; accepted lesson timing.	Economist, curriculum, teacher, architecture review; no unresolved source contradiction blocks the prototype.
1 — Prototype	Your Market, Cap Builders, League Balance vertical slices with all three views and debrief.	Usability with students/adults; reconnect/recovery; honest metrics; first action and room timing targets.
2 — Ramaz Alpha	Six taught lessons mixing prototypes and strongest legacy experiences.	Six session receipts; debrief used; P0/P1 issue log; student explanations and pull; no unsafe data exposure.
3 — Refound	Eight flagship portfolio, shared controls/display, library promotion workflow, Track 101/201 lesson packages.	Twelve additional live runs; stable lesson outcomes; no repeated operator failure; technical SLOs met.
4 — Transfer	Facilitator guides, setup, demo class, support/fallback, non-founder pilot.	Three non-BOW adults run with ≤30 minutes prep; outcome patterns and debrief quality within a defined tolerance; teachers would run again.
5 — Generalize	Selective shared-core extraction and one motif or Financial Literacy primitive pilot.	Second product truly needs the contract; no degradation of Economics; maintenance owner assigned.

Ramaz Alpha lesson set

Lesson	Why included
101 Salary Cap Basics / Cap Builders	Tests Track 101 rules, pair flow, allocation, shock, and option visibility.
101 Small Markets / League Balance	Tests class interdependence, institution design, stakeholder data, rerun.
101 Why Draft Isn't Ranking	Uses a strong existing asset and tests distribution debrief.
201 How Teams Bend the Cap / Cap Crash	Tests dense rule engine, team strategy, recovery, and multi-period shock.
201 Player Economics / Contract Table	Tests private information, bilateral negotiation, bargaining events.
201 Modeling the Market / Model Room	Tests model disagreement, confidence, revision, and explanation.

27. Success metrics and receipts

V5 measures whether the product creates a teachable system, not whether students accumulated points. Metrics are session-level unless a validated study supports a stronger claim.

Dimension	Target / receipt
Entry	Median join under 60 seconds; 95% ready within 2 minutes; no account required for Economics default.
Action	First consequential action within 4 minutes after join; no more than one rules rescue for most teams.
Operability	Freeze acknowledged across clients within 2 seconds; restore/reassign within 30 seconds; Brayden can run without developer help.
Reliability	No unrecovered session loss; disconnect/reconnect and duplicate join logged; invalid rounds visibly marked.
Debrief	Declared metrics available within 10 seconds of close; Brayden can begin a meaningful reveal within 90 seconds.
Learning	Most students can state the mechanism, use one piece of class evidence, and name one boundary or changed condition.
Student pull	Students voluntarily debate strategy or request another run; quick pulse separates fun, clarity, and perceived decision impact.
Teacher transfer	Three non-BOW adults prepare in ≤30 minutes and would run the lesson again.
Portfolio	Every promoted flagship has live receipts, owner, health date, fallback, mechanic distinction, and curriculum job.

Session receipt

Context	Track/lesson; class size; grouping; device/network notes; experience and version.
Run	Timing; controls used; shocks; completion; invalidity; technical recovery.
Pattern	Declared class metrics and graph; what students noticed before explanation.
Reasoning	Prediction change; sample explanations; common misconception; unresolved question.
Pull	Observed energy, confusion, debate, replay request, short student pulse.
Decision	Keep/refine/refound; issue owners; next test; claim boundary.

28. Adversarial product review

The critics below do not certify V5. They define the attacks the build must continue to survive. Where evidence is missing, the answer is a gate—not confidence language.

Critic	Strongest attack	V5 response
Economist	Sports rules can be mistaken for universal economics; class outcomes can be narrated causally.	Declare agents, assumptions, and boundaries; distinguish observed data from model explanation; use neutral mechanisms when sports distorts.
Game designer	Most current experiences are decision wrappers, not systems; a shared shell will not create emergence.	Prototype mechanics with real state, interdependence, tradeoffs, and replay; demote quizzes to supporting activities.
Track 101 student	Rules, math, and dashboards can bury the fun and the decision.	Fewer variables, system-computed arithmetic, pair talk, larger feedback, fast action, concrete consequences.
Track 201 student	Simplified choices can feel childish or have an obvious ‘smart’ answer.	Private information, model disagreement, multi-period state, uncertainty, and context-dependent strategy.
Live teacher	Running devices, a market, questions, and a display can overwhelm one adult.	Stable controls, freeze, automated grouping, recovery, pre-authored shocks, debrief recipes, and Alpha operator metrics.
Brayden facilitator	The product may make Brayden stare at a cockpit instead of the room.	Seven primary controls, display-led room cues, only actionable diagnostics, rehearsal and failure states.
Visual critic	Dark dashboards and animation can masquerade as premium quality.	Role/stakes first, consequence visibility, large class typography, accessibility, and deliberate art direction by motif.
Curriculum critic	Platform ambitions may pull the sequence away from the supplied Track 101/201 curriculum.	Preserve 4×3 structure; separate future opportunities; map every flagship to a lesson job.
Architecture critic	Shared core can become a universal platform that slows both subjects, both arms, and every format.	Small contracts, distinct runtimes, optional bridges, and extraction only after a second real need.
Skeptical product critic	This may be EdTech with sports animation and live graphs.	Student verbs, consequential state, social necessity, model truth, class data, and debrief are promotion gates.

Unresolved risks

Risk	Why it matters	Required test
Founder dependence	A brilliant Brayden-led lesson may not transfer.	Non-founder facilitator test after Ramaz quality is real.
Real-time complexity	Multiplayer and recovery can consume engineering and class time.	Three vertical slices, load/reconnect tests, and strict live-only-when-needed rule.
Student sample	One school and familiar contexts may not generalize.	Record grade, prior knowledge, device, and motif response; later cross-school test.
Sports staleness	Cap rules, deals, and examples change.	Dated instructor notes and versioned data packs.
False evidence	Dashboards can imply learning or causality without support.	Session-bound claim language, explicit data truth states, optional research design.
Library debt	76 records can become a maintenance sink.	Active-shelf limits, owner requirement, automated health, archive aggressively.
Motif ambition	Multiple Worlds can create shallow reskins and content burden.	One cross-motif concept pilot and a World trigger.
Product merger	Shared code can quietly force shared pedagogy.	Architecture review of every shared dependency and product-specific escape path.

The decision after review

PROCEED

Proceed with V5 as a definition-and-prototype program, not a mandate to migrate or build everything. The architecture survives review because it reduces—not expands—near-term scope: three vertical slices, six Ramaz Alpha lessons, eight eventual flagships, a smaller active library, and explicit boundaries around Financial Literacy and Decision Challenges.

29. Decision and hypothesis ledger

Item	Status	Trigger to revisit
Shared core + distinct contracts	DECISION	Repeated duplication or dependency shows the boundary is wrong.
Both subjects may have both program arms	DECISION	A named arm fails a real demand, learning, or operating test—not merely because it differs from the first implementation.
Economics live runtime separate from Challenges	DECISION	No planned merger trigger; revisit only with strong cross-product evidence.
Decision Challenges primarily Financial Literacy × School-Day	DECISION FOR NOW	Two Economics concepts show added independent-application value and a viable evidence contract.

Item	Status	Trigger to revisit
Economics school-day arm	VIABLE HYPOTHESIS	Ramaz lesson quality is proven and a teacher/buyer need justifies standards, transfer, and evidence work.
Financial Literacy enrichment arm	VIABLE HYPOTHESIS	A live or program format is requested and demonstrates value beyond the school-day product.
Later 4×3 curriculum map is canonical	WORKING DECISION	Brayden confirms a newer source or intended lesson change.
Eight flagship portfolio	WORKING SPEC	Ramaz receipts show a different mechanic or lesson deserves promotion.
Your Market first future prototype	WORKING SPEC	A platform risk is better tested by another vertical slice.
Sports flagship	DECISION	Sports blocks comprehension or adoption for a concept.
Creator economy second World	HYPOTHESIS	One shared-mechanic pilot matches learning, pull, and debrief quality.
Teacher transfer	HYPOTHESIS	Three non-BOW adults meet prep/outcome/reuse threshold.
Cross-motif comparability	HYPOTHESIS	Validated comparison-set version supports it.

30. BOW Model V5 on one page

BOW

BOW owns learning through decisions and systems for grades 5–8. Students enter a legible system, make consequential decisions, see feedback, revise, and explain. The identity is shared; the product forms are not.

Subject pillars	Economics and Financial Literacy. Neither subject is defined by one adoption setting or runtime.
Program arms	Each subject may have Enrichment and School-Day arms. Immediate build: Economics × Enrichment at Ramaz. Current Financial Literacy emphasis: School-Day. The opposite two cells are viable, gated paths.
Experience formats	Live Labs, Modules & Guided Practice, and Decision Challenges. The format follows the learning and adoption job; Decision Challenges are not a third subject.
Economics now	A live, facilitated classroom laboratory. Students generate the system data Brayden uses to teach. Immediate focus: exceptional Track 101/201 Ramaz lessons.
Financial Literacy now	School-day modules, guided practice, persistence where useful, and Decision Challenges—while preserving a viable enrichment path.
Decision Challenges	A standalone consequential application-and-evidence format, primarily Financial Literacy × School-Day now. Attempts pin immutable content and evidence versions; teachers retain judgment.
Shared core	Concept canon, content/version records, class/session primitives, privacy/accessibility, design tokens, event envelope, and related educator entry.
Separate ownership	Economics live runtime; Challenge runtime/evidence graph; product-specific teaching and facilitation; product-specific claims.
Build order	Define → prototype three vertical slices → teach six Ramaz Alpha lessons → refound eight flagships → test non-founder transfer → generalize only from evidence.
Portfolio rule	Keep quality, coverage, and receipts; archive count. Nine provisional Keeps, fourteen Refines, ten Refounds, sixteen Merges, four Supporting Activities, twenty-one Archives, two Kills.
North-star lesson	Students predict, act inside a real mechanism, affect one another when appropriate, create class data, see the pattern, change one condition, rerun, and explain why.

NEXT ACTION

Treat Parts I–III as the learning-product foundation. Parts IV–V now add the company, pricing, platform, API, creator, and code-consolidation model required to turn that foundation into a venture-scale system.

PART IV

Company and business model

A venture-scale access company: free learning magic, paid implementation and infrastructure, and a portfolio of sequenced revenue engines.

31. Founder mandate and category decision

COMPANY DECISION

Build BOW as a venture-scale company with an access mission. Strong learning outcomes are the trust constraint. Maximum reach is the distribution engine. Durable enterprise value is what finances the work long enough to matter. BOW may use sponsored and philanthropic capital, but it is not designed as a nonprofit whose survival depends on grants.

The category BOW should own

BOW should lead with one category even though the company eventually combines curriculum, simulations, community, infrastructure, and media. The category is decision learning: structured environments where young people practice consequential choices, receive system feedback, revise, and explain. Economics and Financial Literacy are the first subject pillars. Sports is the first cultural wedge. Middle school is the first beachhead. None of those opening choices defines the ceiling.

Umbrella	The decision-learning platform for young people.
Opening promise	The class becomes the economy: students do not merely read about systems; they act inside them.
First audience	Grades 5–8, beginning with enrichment and middle-school classrooms.
First subjects	Economics and Financial Literacy, with Decision Challenges as an application-and-evidence format.
First motif	Sports business, because it produces immediate roles, stakes, identity, negotiation, uncertainty, and cultural pull.
Long-term ceiling	A school, consumer, enterprise, and infrastructure network through which many people create and participate in decision experiences.

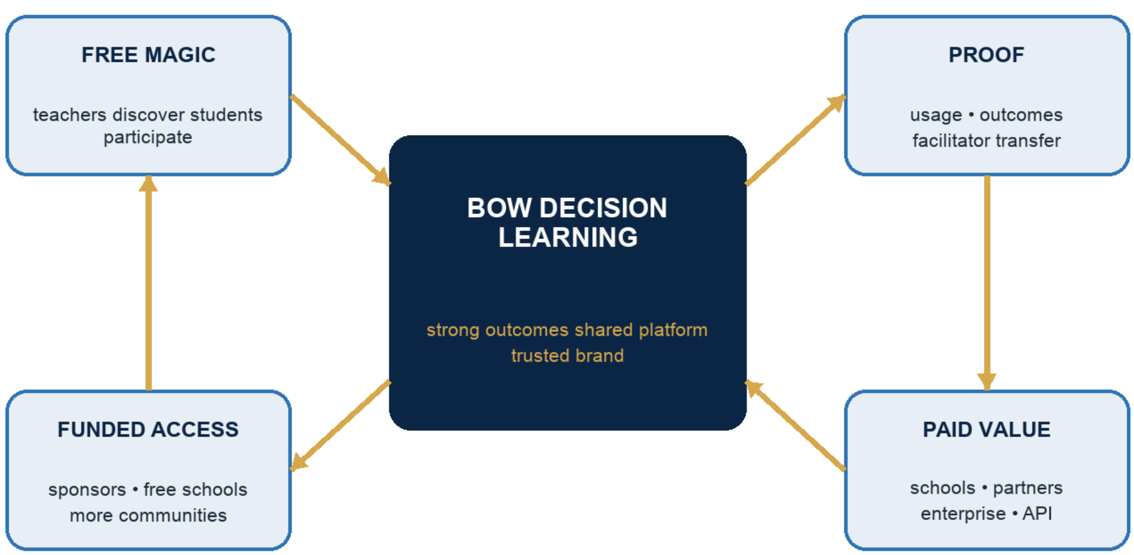
Mission hierarchy

1. Protect learning truth. BOW never trades truthful models, student safety, or genuine decision quality for theatrical growth.
2. Maximize participation. The basic learning magic stays easy to discover, easy to join, and free enough to spread.
3. Build compounding enterprise value. BOW owns reusable technology, a trusted brand, distribution, evidence, creator supply, and partner infrastructure.
4. Use enterprise value to widen access. Sponsors and profitable products finance schools and communities that should not be blocked by budget.

> **OPERATING MAXIM**
>
> No growth that destroys learning trust. No learning work that is designed to remain permanently founder-bound or unscalable.

32. The access-and-scale flywheel

The BOW access-and-scale flywheel



The learner is not the product. Paid layers finance free participation and better learning infrastructure.

Free participation creates adoption and evidence; paid value finances quality, distribution, and sponsored access; better access restarts the loop at greater scale.

The learner is the beneficiary, never the commodity. BOW does not need to charge every learner. It needs to identify which party receives budget-worthy value around that learner: implementation, curriculum, support, reporting, compliance, customization, distribution, engagement, intellectual property, or infrastructure.

Participant	Receives	Pays for
Student	Practice, agency, feedback, identity, and belonging.	Core participation stays free in the reach engine; optional family/League value may be paid later.
Teacher	Fast launch, strong facilitation, debriefs, curriculum, and evidence.	Convenience and depth through Educator Plus; never a toll on trying the core experience.
School/district	Managed implementation, all-staff access, reporting, support, standards, governance, and continuity.	Annual institutional license priced to predictable budget rather than punishing usage.
Sponsor	Measurable community impact and appropriate recognition.	Access Grants that fund schools or regions; never student data or editorial control.
Enterprise/partner	Engagement, branded decision environments, workforce learning, licensed content, or embedded infrastructure.	High-ACV contracts, configuration, licensing, platform minimums, and usage.
Creator	Distribution, tools, trusted standards, analytics, and payment.	BOW retains a transparent marketplace or service share only after the creator system is safe and valuable.

What remains outside the model

- No advertising inside the student learning experience.
- No sale of student data, attention, behavioral profiles, or personally identifiable evidence.
- No gambling, wagering, loot-box economics, pay-to-win mechanics, or speculative tokens.
- No sponsor control over academic conclusions, scoring, standards, or student recommendations.
- No public ranking of children by private financial decisions or evidence traces.

33. What a $10B path actually requires

Five million annual learners is a major mission milestone and a credible route to roughly $100M in annual revenue. It is not, by itself, a dependable $10B-company model. A $10B outcome likely requires tens of millions of active learners, several durable revenue engines, strong retention, global distribution, and roughly $500M or more in annual revenue at a high-growth platform multiple. Valuation multiples are market-dependent and cannot be controlled; the product and revenue system can be.

Scale state	Illustrative reach	Illustrative annual revenue	Meaning
Proof	10,000–100,000 annual learners	$0.5M–$3M	Repeatable learning, transfer, and early paid demand.
Institutional product	0.5M–2M	$10M–$50M	Schools, districts, and sponsors form a real company.
National platform	5M+	$75M–$150M	Major education company; still not automatically a $10B platform.
Category platform	20M–30M+ active learners	$500M–$650M+	Schools, consumer, partners, creators, enterprise, and infrastructure compound.

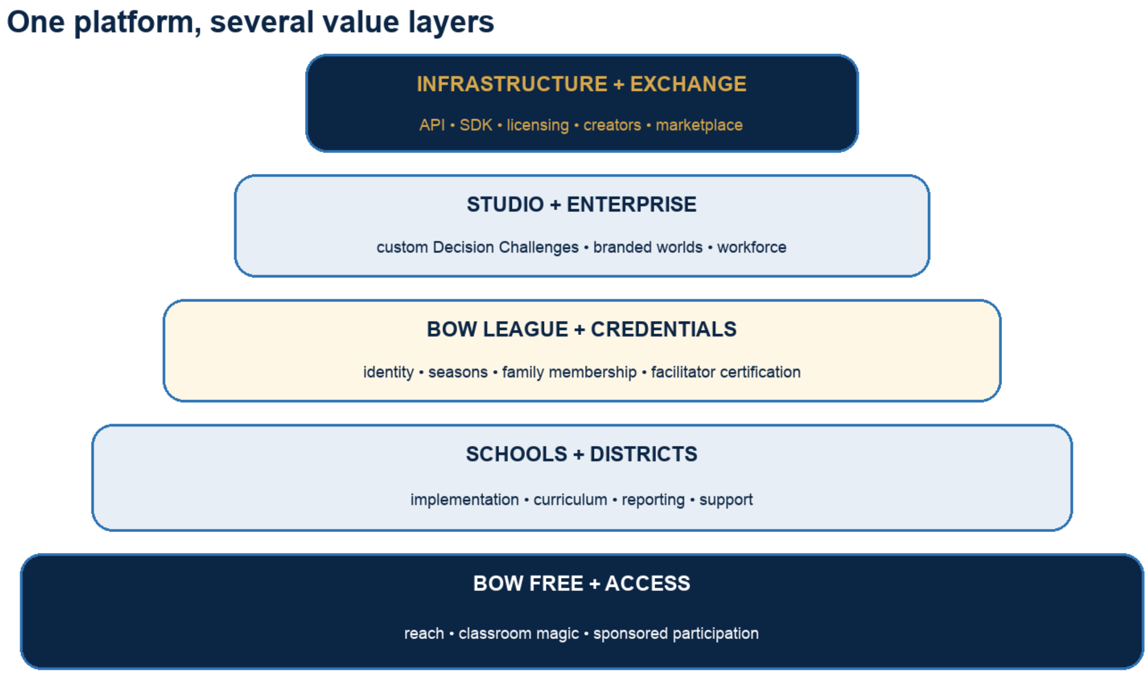
Illustrative category-platform revenue composition

Engine	Scale assumption	Illustrative revenue
Institutional	A blended portfolio of roughly 12,000 paid school equivalents and district agreements.	$180M
BOW League	Approximately 1.5M paying consumer/family members at roughly $90 average annual revenue.	$135M
Infrastructure/licensing	About 200 meaningful platform, publisher, sports, and enterprise partners.	$70M
BOW Access	National, regional, financial-institution, foundation, and employer sponsorships.	$50M
BOW Exchange	A 20% blended BOW share on approximately $300M of creator/content gross volume.	$60M
Studio, certification, events, media	Custom challenges, facilitator network, leagues, camps, and rights.	$75M
Illustrative total	Not a forecast; a test of whether the architecture has a venture-scale ceiling.	$570M

SCALE TEST

At $570M of revenue, a 17.5× revenue multiple would imply approximately $10B. The multiple is not a plan. The useful conclusion is that BOW needs several engines and a global platform ceiling; one middle-school subscription product is not enough.

34. The BOW revenue portfolio



The bottom layer maximizes reach. Higher layers charge for increasing management, persistence, customization, distribution, and infrastructure value.

Engine	Customer	Economic job	Unlock condition
BOW Free	Teachers, programs, students	Near-zero-friction distribution and product-qualified school creation.	Now; only flagships that can be supported honestly.
BOW Schools	Teachers, schools, districts	Predictable recurring institutional revenue.	Non-founder transfer and repeated student evidence.
BOW Access	Banks, teams, employers, foundations	One buyer funds many free participants.	Credible impact reporting and sponsor guardrails.
BOW League	Students and families	Global consumer identity, retention, network, and subscription revenue.	School magic is proven and safe identity/community systems exist.
BOW Studio	Sports, finance, media, workforce	High-ACV custom Decision Challenges and worlds.	Reusable mechanic kernels and delivery discipline.
BOW Infrastructure	Publishers, platforms, partners	API, SDK, runtime, licensing, and white-label revenue.	Two BOW products use stable shared contracts in production.
BOW Exchange	Teachers, studios, publishers	Creator supply, marketplace gross volume, and take rate.	Moderation, certification, IP, payouts, and quality gates work.
BOW Network	Facilitators, camps, leagues, media	Certification, local delivery, events, and rights.	Operating playbooks transfer beyond Brayden.

PORTFOLIO RULE

Every engine must strengthen the same platform. If a proposed business requires a separate identity, separate content truth, separate facilitator system, and separate learning model, it is a new company and should be rejected or explicitly spun out.

35. Subscription and institutional pricing

The core decision is to make learning magic free and managed implementation paid. School pricing should be predictable and should not punish teachers for successful adoption. Usage-based pricing belongs primarily in the partner API, not the school contract.

BOW Free — $0 forever

- Up to 50 participants per live room; one room at a time; unlimited teacher-launched sessions.
- Three strongest flagships: Your Market, Cap Builders, and League Balance Lab, replaced only when another flagship earns promotion.
- Short class code, aliases, no student email requirement, student/player view, facilitator controls, class display, prediction/reveal, core graphs, and debrief.
- Thirty days of aggregate session history, printable facilitator guide, technical fallback, and community support.
- No ads, data sale, artificial session cap, deliberately broken debrief, or deceptive free trial countdown.

Paid plan ladder

Plan	Price	Includes	Boundary
Educator Plus	$199/year or $24.99/month	Up to 100 participants; complete released library; 12-month history; exports; teacher customization; early access; basic portfolios.	One educator; no organization administration, SSO, or district reporting.
Department Launch	$3,500/year	Up to five educators, 300 annual active students, one subject, one professional-learning session.	Maximum two years; then Free, School Complete, or Access Grant.
School Complete ≤500	$7,500/year	All staff and students; both pillars as released; Decision Challenges; portfolios; standards; reporting; support; professional learning.	One school organization.
School Complete 501–1,200	$12,000/year	Same complete institutional entitlement.	One school organization.

Plan	Price	Includes	Boundary
School Complete >1,200	$18,000/year	Same complete institutional entitlement.	One school organization.

District pricing

District size	Annual platform fee	Annual site fee	Example
2–9 schools	$25,000	$7,500/site	Five schools: $62,500.
10–29 schools	$50,000	$6,000/site	Ten schools: $110,000.
30–49 schools	$100,000	$5,000/site	Thirty schools: $250,000.
50+ schools	$100,000+ negotiated	$4,000/site	Scale, support, integrations, and rollout determine final scope.

Billing rules

- Institutional agreements use academic-year annual invoices. Renewal work begins 120 days before expiry; notice begins at 90 days.
- No surprise auto-renewal for institutions. Educator monthly plans may auto-renew with clear self-service cancellation.
- Two-year term discount: 5%. Three-year term discount: 8%. Do not exceed 10% merely for term length.
- Founder paid-school discount: 25% for the first two paid years. Nonprofit discount: 20%. Maximum combined discount: 35%.
- An Educator Plus customer receives prorated credit if the school buys School Complete during the same year.

36. Founding Learning Partners and BOW Access

FOUNDING PARTNER DECISION

Do not charge Ramaz or District 26 during the proof year. Give each a written, time-bounded Access Grant with real entitlements, explicit learning responsibilities, no auto-billing, and no unlimited custom-development promise. Free is the price; the relationship is still a disciplined product partnership.

Partner	Entitlement	Role	Required receipts
Ramaz	School Complete, $0, 2026–27	Invention lab for extraordinary live Economics and founder facilitation.	Student runs, room notes, mechanic evidence, join/recovery results, debrief artifacts, and honest failures.
District 26	District Pilot, $0, 2026–27	Transfer and equity lab for public-school schedules, devices, accessibility, and Financial Literacy fit.	Non-founder facilitation, implementation burden, school constraints, teacher reuse, and access findings.

BOW Access sponsorship menu

Package	Price	Typical footprint	Sponsor receives
Community	$75,000/year	Approximately five schools	Aggregate participation, implementation, and impact reporting; appropriate recognition.
District	$200,000/year	Approximately twenty schools	District access, professional learning, aggregate outcomes, and annual impact story.
Regional	$500,000/year	Approximately fifty schools	Multi-district rollout, implementation team, research-quality aggregate evaluation.
State/major	$1M–$3M+	Large region or statewide initiative	Custom rollout and independent evaluation where appropriate.

- Sponsors receive aggregate impact reporting and recognition; they do not receive student-level data.
- Sponsors cannot require product placement inside a student decision, influence the academic conclusion, or condition access on marketing consent.
- Track the full internal value and cost of every grant so free distribution does not hide an unsustainable delivery model.

37. BOW League and consumer membership

BOW League should not be a homework dashboard with points. It should be the persistent consumer expression of the same decision-learning identity: seasons, clubs, asynchronous and live events, collaborative front offices, identity, progression, and creator experiences. It launches after school sessions have proven student pull and after age-appropriate safety, parental consent, moderation, and account recovery exist.

Plan	Price	Purpose
League Free	$0	Core seasonal participation, public events, a limited profile, and discovery.
League Plus	$8.99/month or $79/year	Expanded seasons, personal history, advanced worlds, cosmetics earned through learning, and deeper analytics.
League Family	$129/year	Multiple child profiles, family challenges, parent controls, and shared events.
League Club	$499/year	Community organizations, clubs, small programs, facilitator tools, and local seasons.

CONSUMER GUARDRAIL

Do not launch BOW League merely to add subscriptions. Launch when the persistent experience is something students voluntarily return to and when retention comes from identity, mastery, creation, and community—not manipulative streak loss or payment pressure.

38. BOW Studio, enterprise, and licensing

Custom Decision Challenges and branded systems can generate the highest early contract values, but custom work becomes dangerous if every engagement creates a one-off runtime. BOW Studio sells configuration of proven mechanic kernels, content production, facilitation, evaluation, and distribution on the shared platform.

Offer	Price guidance	What changes
Enterprise platform	Starting at $50,000/year	Organization, users, reports, content access, support, and governance.
Light configuration	$25,000–$50,000	Brand, numbers, copy, and approved scenario settings on an existing experience.
Scenario on an existing mechanic	$75,000–$150,000	New content package, validation, facilitator materials, and reporting using a proven kernel.
Major new World	$150,000–$500,000+	New institution, interaction system, production, evaluation, and launch—approved only when platform reuse is real.
Content/IP license	$25,000–$150,000/year	Defined subjects, territories, channels, or brands; no transfer of student data.

Out-of-box enterprise paths

- A bank or credit union funds a regional Financial Decisions Season that schools join free.
- A sports team licenses a locally branded front-office World and funds access in its community footprint.

- A publisher embeds BOW Live Labs into an existing economics or financial-literacy curriculum.
- An employer uses adapted Decision Challenges for early-career judgment, resource allocation, negotiation, and risk training.
- A media partner distributes a national Decision League while BOW retains the learning system, accounts, and creator relationships.
- A city or foundation funds civic decision simulations around public goods, budgets, infrastructure, and tradeoffs.

39. BOW Infrastructure and API pricing

API DECISION

Do not price the core partner product by raw HTTP call. The value unit is a learner session, completed challenge, active learner, licensed content package, or governed deployment. API calls are an implementation detail that customers cannot forecast and BOW should not optimize for.

Tier	Price	Included	Overage / expansion
Developer Sandbox	$0	Public catalog, test launches, sample content, non-production keys, and up to 500 test learner sessions/year.	Application required for higher test volume.
Embed	$10,000/year minimum	Signed launch SDK, standard embed, 25,000 learner sessions, aggregate completion webhooks, and email support.	$0.25 per additional learner session.
Platform	$50,000/year minimum	200,000 sessions, SSO/service accounts, content packages, evidence webhooks, analytics, staging, and priority support.	$0.15 per additional session; volume bands negotiated.
Enterprise Infrastructure	$100,000–$300,000+/year	Dedicated governance, advanced controls, regional requirements, service commitments, and partner success.	Usage, licensed content, and support capacity priced in contract.
White label	$150,000–$500,000 setup + annual platform	Approved brand shell, custom domain, implementation, content configuration, and governance.	Annual minimum plus usage or revenue share.
Strategic distribution	Negotiated	Deep platform or media integration where both parties contribute distribution or IP.	Typically 10%–25% revenue share with a guaranteed minimum.

API products

Catalog API	Discover approved experiences, concepts, versions, durations, delivery modes, standards, and launch requirements.
Launch SDK	Create a signed session, apply entitlements, open the correct runtime, and preserve the partner's return path.

Session API	Create, start, pause, close, and recover governed sessions without exposing runtime internals.
Content Packages	License versioned scenarios, numbers, copy, standards, facilitator materials, and permitted brand layers.
Evidence Webhooks	Return completion and approved evidence references; never stream a child's click trail or private decision log by default.
Usage API	Expose auditable learner-session and completion meters for customers and billing.

Use a hybrid commercial model: an annual minimum buys platform readiness and support; included credits make budgeting predictable; a learner-session overage aligns expansion with value. Meter usage inside BOW first, reconcile idempotently, and send validated aggregate meter events to the billing system rather than making the payment provider the source of truth.

40. BOW Exchange, certification, and optional engines

The creator economy is game-changing only after trust exists

Stage	Who creates	BOW control	Economics
Studio	BOW product team	Full design, model, safety, and publication control.	BOW owns revenue and learns the production system.
Certified configurators	Selected teachers/facilitators	Approved kernels and parameter ranges; review before sharing.	Certification fee or included institutional seats.
Invite-only creators	Expert educators and studios	No-code tools, sandbox, automated gates, human publication review.	Creator receives 80%; BOW retains 20% of sales.
Publisher partners	Curriculum and sports/media partners	API contracts, content/version rules, brand and evidence review.	Annual minimum plus usage, license, or revenue share.
BOW Exchange	Qualified global creator network	Certification, moderation, age/IP controls, ratings, refunds, and removal rights.	Standard 20% share; up to 30% when BOW sells, sponsors, or produces materially.

Other viable engines

Engine	Commercial model	When it belongs
Certified Facilitator	$499 individual certification; institutional seats bundled or contracted.	After three non-founder adults can run the product successfully.
Local BOW Labs	Operator license, training, curriculum, and an eventual 5%–10% revenue share.	After camps/clubs have a reproducible operating playbook and child-safety system.

Engine	Commercial model	When it belongs
National Decision League	Sponsors, media rights, premium membership, team entry, and events.	After safe League retention and reliable live-event operations.
Credentials	Included in paid products; verification, advanced pathways, and partner recognition create value.	After credentials represent observed work rather than completion theater.
Research partnerships	Funded evaluation, design partnership, and privacy-governed analysis—not sale of data.	With school approval, minimal datasets, clear retention, and appropriate independent review.
Older learner talent pathways	Opt-in competitions, internships, and recruiting partnerships.	Only for age-appropriate learners with explicit consent; never covert ranking of minors.

DO NOT BUILD ALL OF THIS

The portfolio describes BOW's option value. The company wins by activating one engine only when the previous layer creates the proof, distribution, contracts, or technology the next engine needs.

41. Commercial sequencing and company metrics

Phase	Primary model	Do now	Do not do
Proof: now	BOW Free + Founding Access Grants	Run Ramaz/D26, instrument usage, validate outcomes, test transfer, capture case studies.	Charge pilots, build public marketplace, promise statewide results, or add consumer subscriptions.
Early revenue	Educator Plus + Department + School Complete	Convert product-qualified schools, deliver professional learning, establish renewal behavior.	Custom-build for every buyer or sell integrations before core adoption works.
Distribution	District + BOW Access + Studio	Fund free footprints, standardize implementation, sell reusable custom work.	Let sponsors direct curriculum or create permanent services burden.
Network	League + Infrastructure + Certification	Extend identity beyond school, license stable components, train delivery network.	Open unmoderated child community or price APIs by confusing raw calls.
Platform	Exchange + media + global partners	Let qualified creators and distributors compound supply and reach.	Relax model truth, safety, evidence, or IP gates for volume.

Product-qualified school trigger

PQS

A school becomes product-qualified when it reaches at least three active teachers, 100 student joins, five sessions, two simulations, and activity in two separate months. Then offer a school usage summary, a School Complete demonstration, a 60-day institutional pilot, and a written proposal.

Metrics that matter

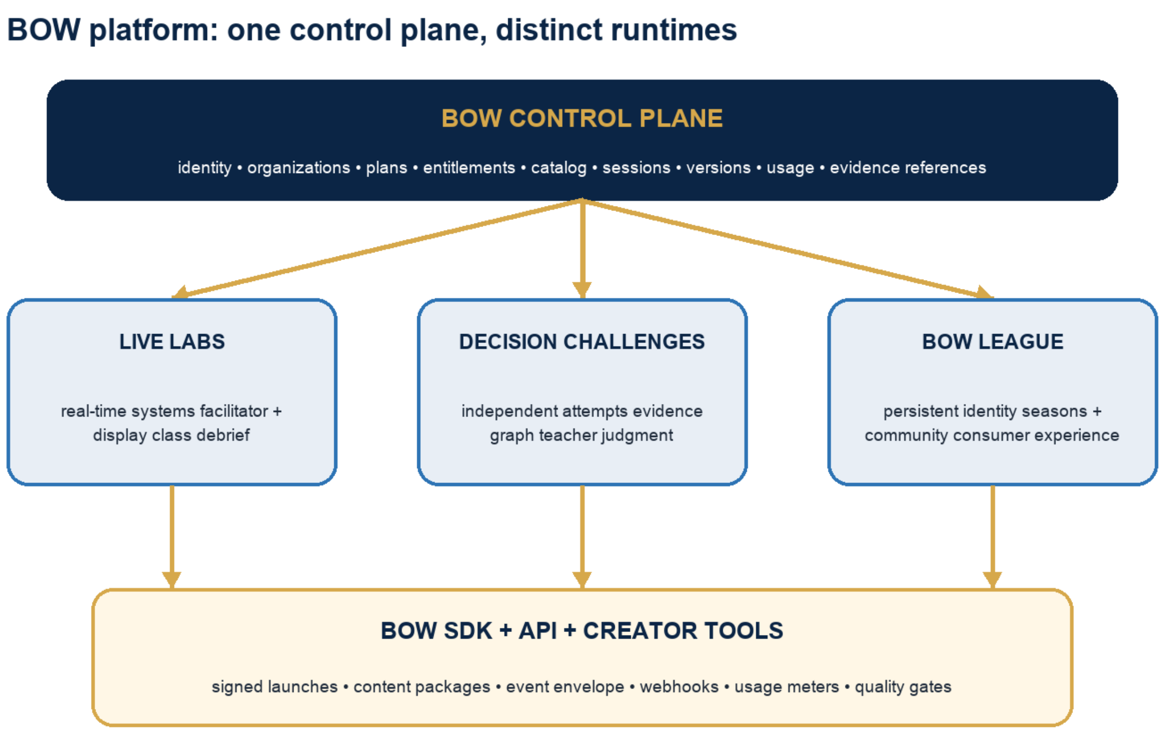
Layer	Primary metrics
Learning	Decision quality, revision quality, concept explanation, misconception movement, transfer tasks, and teacher judgment.
Reach	Annual active learners, active teachers, participating schools, free-to-paid distribution, sponsored access, and geographic/need coverage.
Product	Join time, completion, replay, facilitator intervention, technical recovery, student pull, and non-founder success.
Revenue	ARR, gross retention, net revenue retention, expansion, acquisition cost, implementation cost, gross margin, and revenue concentration.
Platform	Experiences per kernel, partner launches, creator activation, API session volume, approval time, content health, and marketplace gross volume.
Mission integrity	No data sale, privacy incidents, sponsor exceptions, accessibility defects, unsafe community events, or misleading evidence claims.

PART V

Platform and simulation system

Turn the current repository portfolio into one company: a canonical control plane, distinct runtimes, shared contracts, and an evidence-gated simulation factory.

42. The platform decision



The control plane owns company-wide truth. Runtimes own the interaction models that make their learning experiences excellent.

TECHNICAL DECISION

Make BOW-WEBSITE the canonical control plane and public front door. Do not force every experience into its Next.js codebase. Connect Live Labs, Decision Challenges, and later BOW League through signed launches, shared contracts, event/evidence references, and entitlements.

Control plane ownership

Control plane owns	Runtime owns
User and pseudonymous participant identities; organizations; memberships; roles; plans; subscriptions; Access Grants; sponsor grants; entitlements; content catalog; versions; session index; usage records; approved evidence references; support and audit.	Interaction state; economic or financial engine; timers; private information; scoring or evidence derivation; facilitator-specific controls; runtime recovery; presentation; detailed event vocabulary.

A session starts in the control plane, which checks entitlement and produces a short-lived signed launch. The selected runtime validates the launch, runs the experience, writes its own detailed state, and returns a standardized completion, usage, and evidence summary. The control plane never needs to understand every bid, contract, portfolio decision, or rubric event.

43. One BOW identity with three participation modes

Mode	Use	Identity	Persistence
Guest session	Default live classroom and free discovery	Short code + safe alias; no email.	Session token; minimal retention; optional teacher-approved handoff.
School-managed learner	Programs, portfolios, Decision Challenges	School organization membership or roster-linked pseudonymous identity.	School-governed history, evidence, and deletion/retention rules.
Consumer member	BOW League, family, clubs, public seasons	Age-appropriate account with required parent/guardian and regional controls.	Portable identity, achievements, creations, and consent-managed profile.

- Never require an email for a child to join an ordinary live classroom session.
- Do not silently merge a guest alias with a permanent account. The learner or school completes an explicit handoff.
- Keep school evidence and consumer community data in separate policy domains even if one person later links them.
- Default big-screen and sponsor reporting to anonymous aggregates.
- Every stored record has a purpose, owner, retention rule, access rule, and deletion path.

44. GitHub portfolio disposition

The account contains 83 repositories and the Simulation Library identifies 76 experiences. This is valuable product exploration, not a scalable production topology. The goal is to consolidate company truth while preserving successful mechanics and evidence. Count is not an asset if users cannot find, launch, trust, or reuse the products.

Repository/family	Decision	New job
BOW-WEBSITE	KEEP + ELEVATE	Canonical public site and BOW control plane: identity, organizations, plans, entitlements, programs, catalog sync, sessions, usage, support, and institutional reporting.
sim-library	KEEP INDEPENDENT	Canonical product registry, provenance, health, validation, curation, public-readiness, and promotion state. It does not become the runtime.
bow-finlit	KEEP AS RUNTIME	Front Office Challenge and the strongest live-class/session behavior. Extract contracts and reusable session primitives only after they prove a second use.
bow-decision-challenges	KEEP AS RUNTIME	Canonical independent challenge, immutable evidence, world-neutral engine, and teacher-reading surface.
bow-universe	INCUBATE	Future BOW League research environment. Preserve proposals, missions, community, and persistent-world learnings; do not make it the current school core.
Bow-Sports-Capital-Full-APP	MINE + SUNSET	Harvest credentials, quests, passes, pods, seasons, and iframe-launch ideas into the control plane or League roadmap; stop parallel identity/billing architecture.
Bow-Platform / Apps Script	MIGRATE + ARCHIVE	Move still-used operational records and workflows into BOW-WEBSITE; freeze as a read-only historical adapter after reconciliation.
Individual lesson/sim repos	MANIFEST + CURATE	Remain independent prototypes until promoted. Add a manifest, verified launch, ownership, version, event/usage adapter, and archive rule.
league-in-a-box and small shells	MECHANIC SOURCE	Preserve distinctive mechanics or content only; do not promote template repositories into production applications.
Sports analytics models / CourtIQ	OPTION PORTFOLIO	Advanced curriculum, older-learner, enterprise, or licensing R&D; separate from the middle-school control plane until a customer and learning job exist.

What not to do

- Do not rewrite 76 experiences into one framework before eight flagships are proven.
- Do not run multiple production identity, organization, billing, entitlement, and portfolio systems.
- Do not choose between Prisma, Drizzle, and direct SQL by ideology. Standardize the control plane on BOW-WEBSITE’s Postgres migration path; let runtimes keep local internals until migration creates more value than risk.
- Do not delete historical repositories. Mark superseded products, preserve provenance, and archive them after their reusable assets are named.
- Do not let a repository’s default branch remain an experimental agent branch once the product becomes production-critical.

45. Platform records and entitlement architecture

Record	Purpose
User	An adult or persistent learner identity; not required for a guest participant.
Participant	A session-scoped or linked learner actor with a safe display identity.
Organization	School, district, sponsor, enterprise, club, household, or internal BOW organization.
Membership	User-to-organization role and lifecycle.
Product	Commercial or free product family: Free, Schools, League, Studio, Infrastructure.
Plan	Named packaging and public/commercial rules.
Subscription	Paid relationship, dates, billing state, quantity, and renewal.
AccessGrant	Time-bounded no-charge entitlement such as Ramaz or D26; separate from subscription.
SponsorGrant	Funding source, funded footprint, restrictions, aggregate reporting, and expiry.
Entitlement	Machine-readable capability independent of price: participant cap, content, history, reporting, integrations, API.
ExperienceVersion	Immutable approved version of a simulation, challenge, World, content package, and evidence contract.
Session	Control-plane index for a runtime run, organization, facilitator, version, status, and signed launch.
UsageRecord	Idempotent commercial unit such as learner session started/completed, report generated, or API package used.
EvidenceReference	Approved pointer and summary; detailed evidence remains governed by the runtime and school policy.

SEPARATION RULE

Plan describes packaging. Subscription describes a paid relationship. Access Grant describes free authority. Sponsor Grant describes who funded access. Entitlement describes what the software permits. Never collapse these into one subscription tier field.

Minimum shared contracts

Contract	Required truth
SimulationManifest	Stable ID, approved version, subject, concept links, runtime URL, delivery, duration, privacy/data profile, evidence output, entitlement requirement, health, and owner.
LaunchContract	Organization, facilitator, participant mode, experience version, allowed capabilities, expiry, return path, and signed nonce.
SessionContract	Status vocabulary, participant cap, runtime ownership, recovery identifier, start/end, and usage policy.
ControlContract	Which actions the facilitator may invoke and what each action truthfully changes.

Contract	Required truth
EventEnvelope	Version, session, actor role, event type, payload, sequence, timestamp, support level, and technical status.
EvidenceContract	What evidence exists, what it can support, derivation version, visibility, retention, and human-judgment boundary.
DebriefRecipe	Declared metrics, thresholds, graphs, comparisons, model explanation, and claim limits.
UsageMeter	Commercial unit, idempotency key, organization, product, quantity, timestamp, source version, and reconciliation state.

46. The simulation portfolio BOW should build

BOW should use a barbell portfolio: a small free shelf of fast, excellent discovery experiences and a smaller number of deep flagships that create learning, retention, evidence, and paid institutional value. The company does not need hundreds of equally promoted simulations.

Format	Length	Job	Business role
Spark	5–15 minutes	One consequential decision and immediate interpretable feedback.	Free discovery, sharing, teacher acquisition, event hooks.
Live Lab	35–60 minutes	Students form a social system; facilitator uses class outcomes to teach and rerun.	BOW Free magic and School Complete differentiation.
Decision Challenge	One to four sessions	Independent application, revision, evidence, and teacher judgment.	School-day depth, enterprise configuration, credentials.
Season	Days to months	Persistent teams, proposals, markets, events, identity, and community.	BOW League retention, membership, sponsors, media.
Kernel/Kit	Reusable	Validated economic/financial engine, controls, content schema, and debrief system.	Studio margin, API licensing, creators, marketplace.

Portfolio allocation for the next two years

- Build or refound six to eight extraordinary flagships before expanding the visible shelf.
- Use sports for approximately 70% of near-term flagship production because it is the brand wedge and richest existing asset base.
- Use everyday personal finance for approximately 20% so Financial Literacy is a true co-equal pillar rather than a promise.
- Use approximately 10% for one alternate motif—creator economy is the leading candidate—to test whether BOW’s mechanics travel beyond sports.
- Every flagship should produce one or more Sparks, one facilitator-ready core experience, and a reusable mechanic or content package where honest.

QUALITY DECISION

Prefer twelve simulations students remember, teachers rerun, partners can license, and creators can extend over 100 simulations that exist only as repository links.

47. The simulation factory

Every simulation moves through the same evidence gates even when its interaction model differs. Production speed comes from reusable kernels, contracts, test harnesses, content schemas, design primitives, and facilitation patterns—not from making every experience the same game.

1. Name the learning job. State the concept, misconception, decision, consequence, revision, and evidence claim in plain language.
2. Choose the smallest honest mechanic. Define agents, resources, information, actions, state transition, uncertainty, and model boundary before designing screens.
3. Prototype the decision loop. Use a disposable prototype to test whether students face a real tradeoff and whether different reasonable strategies can exist.
4. Run the design-time model. Simulate many futures, test degenerate strategies, inspect distributions, and prove the scoring/debrief does not reward unintended behavior.
5. Build one vertical slice. Student view, facilitator or independent flow, consequence, recovery, event envelope, and debrief/evidence must all work together.
6. Pass the truth gates. Economist/subject review, accessibility, privacy, safety, content/IP, technical reliability, and honest evidence language.
7. Test with students. Observe comprehension, pull, reasoning, time, device failure, and unintended strategies; record the exact version.
8. Test transfer. A non-founder facilitator prepares, launches, runs, recovers, and debriefs without Brayden silently rescuing the session.
9. Promote or archive. CORE requires repeated use and transfer. A beautiful prototype that fails the room returns to R&D or is archived.
10. Extract only proven reuse. A mechanic becomes a kernel after at least two successful experiences need the same engine and their differences can stay content-level.

Flagship publication gate

Gate	Must be true
Decision	A student commits under a meaningful constraint before knowing the consequence.

Gate	Must be true
System	The result comes from declared state, interaction, uncertainty, or rules—not an arbitrary answer key.
Learning	The experience makes a named concept or misconception more visible than direct explanation alone.
Fun/pull	Students understand the goal, want to act, and have a reason to compare, revise, or replay.
Facilitation	An adult can run or assign it with bounded preparation and recover from ordinary failure.
Truth	Model assumptions, limits, evidence claims, and scoring are documented and tested.
Safety/access	Age, privacy, accessibility, display, identity, and content/IP requirements pass.
Technical	Launch, reconnect/resume, version pinning, recovery, telemetry minimum, and fallback work.
Commercial	Entitlement, usage unit, support owner, maintenance cost, and intended free/paid role are explicit.

48. Creator tools without losing BOW quality

Teacher and creator participation can become BOW's strongest supply-side moat, but open authoring is not the first product. The first creator tool should be a Scenario Studio that lets qualified adults change safe variables and content inside a proven kernel—not a blank canvas that creates unreviewed learning claims.

Creator capability	Unlock
Parameter presets	A BOW builder changes approved numbers, shocks, timing, role counts, and comparison conditions.
Scenario Studio	Certified facilitators author copy, roles, numbers, debrief prompts, and standards on a proven kernel; automated gates run before preview.
World Studio	Invite-only experts create coherent scenario packs and art direction while engines and evidence remain governed.
Partner SDK	Approved publishers and organizations create content packages, launch sessions, and receive evidence webhooks.
Exchange publication	Human review, rights verification, model/evidence review, age rating, accessibility, support owner, pricing, refund, and removal policy.

CREATOR PRINCIPLE

Creators may own expression, scenarios, pedagogy, and distribution. BOW must retain the right to enforce model truth, safety, privacy, accessibility, versioning, evidence language, IP, and publication quality.

49. Exact build sequence from the current code

First 90 days

Window	Build	Proof
Days 1–15	Declare BOW-WEBSITE the control plane. Add Product, Plan, Entitlement, AccessGrant, SponsorGrant, ExperienceVersion, Session, Participant, UsageRecord, and EvidenceReference designs. Freeze creation of competing identity/billing systems.	Architecture decision record, schema plan, repo ownership list, Ramaz/D26 grant records.
Days 16–30	Import/sync the Simulation Library as catalog truth. Add entitlement checks and a signed launch contract. Register bow-finlit and Decision Challenges as external runtimes.	One catalog record launches each runtime through the control plane in a test environment.
Days 31–45	Implement guest-session identity, organization context, session index, idempotent learner-session usage, return webhooks, and safe aggregate completion.	A guest joins without email; BOW records one truthful usage unit; duplicate webhook does not duplicate usage.
Days 46–60	Refound the first Economics vertical slice—Your Market—using the live contract, class display, freeze/recover, debrief recipe, and versioned manifest.	A simulated class and internal rehearsal pass launch, action, recovery, graph, rerun, and close.
Days 61–75	Run Ramaz pilot sessions, instrument join/recovery/facilitation, capture student reasoning, and repair the highest-friction failures.	Exact tested versions and operator receipts enter the Simulation Library governance record.
Days 76–90	Prepare District 26 transfer: facilitator guide, device fallback, teacher rehearsal, Financial Literacy selection, privacy approval packet, and post-session review.	A non-founder adult completes rehearsal without Brayden operating the product.

Months 4–12

1. Complete six validated flagships across Economics and Financial Literacy; retire or hide unmaintained public links.
2. Run each flagship repeatedly, not once; require exact-version student receipts and at least three non-founder facilitators across the portfolio.
3. Launch BOW Free with three flagships and product-qualified-school usage summaries.
4. Add Educator Plus and one School Complete institutional pilot only after transfer works.
5. Sell the first Access sponsor only with a defined footprint, aggregate reporting, and implementation capacity.
6. Keep BOW League, marketplace, white label, and public API in design/prototype until school adoption and shared contracts are proven.

50. Twelve- and twenty-four-month scorecards

Horizon	Product and learning	Reach and transfer	Commercial
12 months	Six student-tested flagships; repeated exact-version runs; two pillars represented; join and recovery targets met; learning/evidence claims calibrated.	10,000 learner sessions; 25 active educators; at least three successful non-founder facilitators; Ramaz and D26 case evidence.	$250K–$750K booked from early schools, sponsorship, or Studio after proof; renewal pipeline visible; no forced pilot conversion.
24 months	Twelve validated flagships; three reusable kernels; school and enrichment pathways; one safe alternate motif; reliable platform launch/evidence contracts.	100,000 annual learners; 100 active schools or equivalent footprint; 20 transfer-ready facilitators; meaningful underserved access.	$1M–$3M ARR; multiple revenue sources; first district/sponsor renewals; one infrastructure or licensing partner; healthy implementation margin.

Founder role

BRAYDEN'S JOB

Remain deeply involved as founder, public teacher, product visionary, and chief standard-setter. Use personal facilitation to invent and understand the magic—but treat every founder-led success as incomplete until the system records why it worked and another adult can reproduce the intended learning. Brayden should become more leveraged, not less involved.

Keep	Teaching flagship sessions; observing students; setting product taste; telling the public story; selecting strategic partners; protecting learning truth.
Systematize	Session playbooks, product reviews, content/version ownership, partner qualification, pricing authority, pilot charters, and evidence review.
Delegate first	Routine support, deployment, data cleanup, scheduling, standard professional learning, invoice operations, and repetitive content conversion.
Never become	The only facilitator, only deployer, only sales channel, only person who understands pricing, or permanent integration layer among 83 repositories.

51. External reference points and limits

These reference points do not prove BOW's model. They show that the components—freemium scale, sponsored school access, creator ecosystems, and hybrid usage billing—exist in large adjacent businesses. BOW must still earn its own learning quality, retention, distribution, and economics.

Duolingo 2025 Form 10-K: https://www.sec.gov/Archives/edgar/data/1562088/000162828026012494/duol-20251231.htm — Reference for the gap between active reach, paid subscribers, and billion-dollar consumer education revenue.

Roblox 2025 Form 10-K: https://www.sec.gov/Archives/edgar/data/1315098/000131509826000024/rblx-20251231.htm — Reference for global active participation, a paying minority, platform infrastructure, and creator economics at very large scale.

Epic engagement payouts: https://dev.epicgames.com/documentation/fortnite/engagement-payout-in-fortnite-creative — Reference for an ecosystem that shares platform revenue with creators based on the value their experiences add.

Stripe usage-based billing: https://docs.stripe.com/billing/subscriptions/usage-based/how-it-works — Reference for annual/recurring relationships combined with auditable usage meters and alerts.

EVERFI K–12 sponsorship model: https://everfi.com/wp-content/uploads/2026/02/Financial-Education-Sponsorship-One-Pager.pdf — Reference for partners funding free school access to financial education.

Council for Economic Education 2026 Survey of the States: https://www.councilforeconed.org/wp-content/uploads/02-24-26_CEE_2026Brochure_2fxgrx_oy-2.pdf — Reference for the current policy and teacher-distribution environment around economics and personal finance.

FINAL COMPANY DESIGN

BOW begins as a middle-school, sports-powered decision-learning product. It scales by making the basic magic free, making schools pay for managed implementation, letting sponsors fund access, building a persistent consumer League, licensing proven infrastructure, and eventually allowing qualified creators to expand supply. The company remains one platform because identity, trust, content truth, entitlements, and distribution compound across every layer.

APPENDIX A

Complete simulation audit

All 76 canonical records, classified for the V5 portfolio. A Keep is provisional until live classroom validation.

A.1 — KEEP

9 records in this decision class.

Cap Crash

Canonical ID	cap-crash
Track / slot	201 • 201 · M1 L1
Current system	track-201 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	allocation-under-constraint, shock-and-adapt • allocate, build, respond, revise
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/T201-M1-L1 • https://braydenokley13-ux.github.io/T201-M1-L1/
Recorded finding	A genuinely open sandbox with a non-obvious result: buying the best affordable players produced four players and 37 wins, buying value produced thirteen and 41. The roster-size rule closes the star-stacking exploit by construction, and season variance is separated from the choices.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

Draft Room Challenge — Fit vs BPA

Canonical ID	draft-room-fit-vs-bpa
Track / slot	101 • 101 · M4 L3
Current system	track-101 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	draft-and-selection • evaluate, select
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/101-M4-L3 • https://braydenokley13-ux.github.io/101-M4-L3/
Recorded finding	The cleanest tradeoff design measured. Always taking the best available player scored 100 and always taking team fit scored 775, yet a rebuilding team rewards the BPA pick that fit-first gets wrong, so no single heuristic survives the scenarios. No defects found.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

Mega City Tycoon

Canonical ID	mega-city-tycoon
Track / slot	101 • Not canonically mapped in the source curriculum
Current system	track-101 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	allocation-under-constraint, business-operator • allocate, build, operate, invest
Delivery / evidence	individual • facilitator any-instructor • evidence none

Source	braydenokley13-ux/101-M1-ECON • https://braydenokley13-ux.github.io/101-M1-ECON/
Recorded finding	Civic economics done properly for the youngest audience: the budget constraint is genuinely enforced rather than decorative, and the closing reflection question is generated from the projects the student actually skipped.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

MLB Money Maker

Canonical ID	mlb-money-maker
Track / slot	201 • 201 · M2 extension
Current system	track-201 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	negotiation • negotiate
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/T201-M2-L1 • https://braydenokley13-ux.github.io/T201-M2-L1/
Recorded finding	Multi-stakeholder bargaining that cannot be solved by one lever. Every allocation tried left someone under 55 percent happy, an even split still collapses the deal, and the second phase forces non-monetary terms. Both the win and the collapse were reached, and no defects were found.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

MLB Player Economics — Agent Simulator

Canonical ID	mlb-agent-simulator
Track / slot	201 • 201 · M2 L2
Current system	track-201 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	negotiation, investment-and-portfolio • negotiate, invest, allocate
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/201-M2-L2 • https://braydenokley13-ux.github.io/201-M2-L2/
Recorded finding	The clearest expected-value teaching anywhere in the account — the EV edge is shown before the choice and each result names the counterfactual not taken. Three arithmetic defects and a state bug were repaired in this pass; what remains is a balance question about the contract data.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

Small Markets, Big Money

Canonical ID	small-markets-big-money
Track / slot	101 • 101 · M2 L2

Current system	track-101 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	allocation-under-constraint • allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/101-M2-L2 • https://braydenokley13-ux.github.io/101-M2-L2/
Recorded finding	The best writing in the account, over real branching. Every option shows its stakeholder deltas before commitment, and two opposed paths ended far apart on competitive balance. Nothing here can be bluffed.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

The Front Office Dashboard — Data to Decisions

Canonical ID	front-office-dashboard
Track / slot	201 • 201 · M3 L1
Current system	track-201 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	draft-and-selection, allocation-under-constraint • evaluate, select, allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/201-M3-L1 • https://braydenokley13-ux.github.io/201-M3-L1/
Recorded finding	The best decision design in the account. TRAP and ALBATROSS player tiers exist specifically to punish buying famous names, a randomised owner mandate changes the constraint every run, and a season simulation with a trade deadline forces adaptation after the roster is set. Played clean with no exploit found.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

Trade Deadline War Room (Track 201)

Canonical ID	trade-deadline-war-room-201
Track / slot	201 • 201 · M4 L3
Current system	track-201 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	trade-and-exchange, shock-and-adapt • trade, respond, revise
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/T201-M1-L2 • https://braydenokley13-ux.github.io/T201-M1-L2/
Recorded finding	Real luxury-tax arithmetic where the same trade is right or wrong depending on the team: the identical star swing produced a 209 million dollar tax bill and an alarmed owner on one roster and a zero bill and a delighted one on another. No defects, no exploit, and the best mobile layout tested.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

Why the Draft Isn't a Ranking

Canonical ID	why-the-draft-isnt-a-ranking
Track / slot	101 • 101 · M4 L1
Current system	track-101 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	draft-and-selection • evaluate, select
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/101-M4-L1 • https://braydenokley13-ux.github.io/101-M4-L1/
Recorded finding	The only simulation that proves its own thesis instead of asserting it: replaying one strategy twenty times returns scores from 40 to 93, and process and outcome are reported separately. The phone layout and keyboard access were repaired in this pass.
V5 decision	KEEP. Preserve the decision core; run live before promotion and change only against evidence.

A.2 — REFINE

14 records in this decision class.

Deal Dynasty — Business Strategy Game

Canonical ID	deal-dynasty
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	negotiation, auction-and-bidding • negotiate, bid
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/301-M3-ECON • https://braydenokley13-ux.github.io/301-M3-ECON/
Recorded finding	Signalling, BATNA, repeated games and multi-party bargaining in one coherent arc, with feedback that explains why an option failed. Reputation currently carries no weight at the ending.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Decision Room Simulator

Canonical ID	decision-room-simulator
Track / slot	301 • 201 · M3 L2 extension
Current system	track-301 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	signal-vs-noise, crisis-management • analyze, forecast, prioritize, respond
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/301-M4-L2 • https://braydenokley13-ux.github.io/301-M4-L2/
Recorded finding	Teaches epistemics rather than arithmetic: two model dashboards disagree, each with a named blind spot, and a mid-run injury event breaks whichever story the student had settled on. Distinct play styles produce distinct archetypes rather than better or worse scores.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

ESPN Crisis Manager: The Salary Cap Leak

Canonical ID	espn-crisis-manager
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	crisis-management • prioritize, respond
Delivery / evidence	individual • facilitator any-instructor • evidence none

Source	braydenokley13-ux/301-M2-L1 • https://braydenokley13-ux.github.io/301-M2-L1/
Recorded finding	The most dramatic consequence divergence measured — trust 0 and revenue 0 against a 95 out of 100 run — and excellent writing. Clicking the first option every round scores 95 without reading, which shuffling option order would fix.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Front Office Draft

Canonical ID	front-office-draft
Track / slot	201 • 201 · M3 L3
Current system	track-201 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	allocation-under-constraint • allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence completion
Source	braydenokley13-ux/201-M3-L3 • https://braydenokley13-ux.github.io/201-M3-L3/
Recorded finding	A complementarity puzzle with real synergy and conflict penalties and four hidden combinations to find. Two different numbers are labelled Final Score on the same results screen.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Front Office: Build the Roster

Canonical ID	front-office-build-the-roster
Track / slot	101 • 101 · M1 L1
Current system	track-101 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	allocation-under-constraint, shock-and-adapt • allocate, build, respond, revise
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/T101-M1-L1 • https://braydenokley13-ux.github.io/T101-M1-L1/
Recorded finding	A randomised owner curveball and a real failure state — a deliberately poor build scored 51 against 99. The economics lean on soft chemistry and clout rather than cap arithmetic.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Front Office: The Homestand

Canonical ID	front-office-the-homestand
Track / slot	101 • 101 · M2 L1

Current system	track-101 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	market-and-pricing, business-operator • price, buy, sell, operate, invest
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/T101-M2-L1 • https://braydenokley13-ux.github.io/T101-M2-L1/
Recorded finding	Three chained decisions with consequences narrated clearly and real divergence between paths. Deterministic and unscored, which suits discussion and blunts replay.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Gauntlet L2: Supply Chain Crisis

Canonical ID	gauntlet-l2-supply-chain-crisis
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	gauntlet • active · PLAYABLE · healthy • curation NONE
Mechanic / verbs	business-operator, market-and-pricing • operate, invest, price, buy, sell
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/GAUNTLET • https://braydenokley13-ux.github.io/GAUNTLET/GauntletL2/
Recorded finding	Reachable, and the only Gauntlet level with a live results endpoint wired in. Probed 2026-08-14.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Moneyball Draft Challenge

Canonical ID	moneyball-draft-challenge
Track / slot	101 • 101 · M3 L1
Current system	track-101 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	draft-and-selection • evaluate, select
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/101-M3-L1 • https://braydenokley13-ux.github.io/101-M3-L1/
Recorded finding	A tight constrained-optimisation puzzle that is gated rather than narrated: chasing the flashy players is rejected outright and does not advance.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

NBA Surplus Value Championship

Canonical ID	nba-surplus-value-championship
Track / slot	201 • 201 · M4 L2
Current system	track-201 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	draft-and-selection • evaluate, select
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/201-M4-L2 • https://braydenokley13-ux.github.io/201-M4-L2/
Recorded finding	Real multi-objective tension between surplus and synergy. Held back by a win check that does not match the target printed on screen, which is a one-line fix in a simulation about reading your own numbers.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Risk, Volatility & Rational Aggression

Canonical ID	risk-volatility-gm-sim
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	business-operator, investment-and-portfolio • operate, invest, allocate
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/301-M2-L2 • https://braydenokley13-ux.github.io/301-M2-L2/
Recorded finding	The richest simulation in the account: stochastic seasons, enforced cap constraints, and scoring that reads the moves actually made rather than the strategy label chosen. The dashboard that told four of five team contexts to play it safe — costing a student up to 32 points of 100 for following it — was repaired in this pass.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Stats vs Scouts — Basketball Decision Lab

Canonical ID	stats-vs-scouts
Track / slot	101 • 101 · M3 L2
Current system	track-101 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	signal-vs-noise, draft-and-selection • analyze, forecast, evaluate, select
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/101-M3-L2 • https://braydenokley13-ux.github.io/101-M3-L2/

Recorded finding	Real historical cases that resist a single heuristic, plus a genuine value-for-money round. The final round looks frozen while a hidden timer runs out.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

The GM Challenge — Master the Efficiency Frontier

Canonical ID	gm-challenge-efficiency-frontier
Track / slot	301 • 201 · M3 extension
Current system	track-301 • active · PLAYABLE · healthy • curation FLAGSHIP
Mechanic / verbs	draft-and-selection, allocation-under-constraint • evaluate, select, allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/301-M1-L3 • https://braydenokley13-ux.github.io/301-M1-L3/
Recorded finding	The most numerically honest marginal-analysis lesson available. The same five players bought at premium rather than efficient prices cost 53 percent more for 24 percent more wins, and efficiency visibly falls from 48 to 39 percent.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Trade Deadline War Room (Track 101)

Canonical ID	trade-deadline-war-room-101
Track / slot	101 • 101 · M4 L3
Current system	track-101 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	trade-and-exchange, shock-and-adapt • trade, respond, revise
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/T101-M1-L2 • https://braydenokley13-ux.github.io/T101-M1-L2/
Recorded finding	Strong production and a genuine ownership pressure beat, undermined by scoring that rates standing pat twice above the most aggressive path available.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

Venture Capital Tycoon

Canonical ID	venture-capital-tycoon
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation RECOMMENDED

Mechanic / verbs	investment-and-portfolio • invest, allocate
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/301-M1-ECON • https://braydenokley13-ux.github.io/301-M1-ECON/
Recorded finding	Five years of portfolio construction where passing on everything is correctly punished. A visible impossible equity figure and templated feedback that contradicts its own scorecard need fixing before it is used for assessment.
V5 decision	REFINE. Target the recorded defect, balance, facilitation, data, or interaction weakness; do not rebuild the premise unnecessarily.

A.3 — REFOUND

10 records in this decision class.

BOW Boss Sim — Economic Summit

Canonical ID	bow-boss-sim-economic-summit
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	gauntlet • active · PLAYABLE · broken • curation HOLD
Mechanic / verbs	negotiation, allocation-under-constraint • negotiate, allocate, build
Delivery / evidence	pairs • facilitator any-instructor • evidence completion
Source	braydenokley13-ux/GAUNTLET • https://braydenokley13-ux.github.io/GAUNTLET/Boss%20Sim/
Recorded finding	The strongest negotiation premise in the account, and unplayable: no honest run gets past round 2 of 6 because the continue control is never rendered.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

BOW Sports Capital — GM Decision Game

Canonical ID	gm-decision-game
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation REBUILD
Mechanic / verbs	business-operator • operate, invest
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/301-M1-L1 • https://braydenokley13-ux.github.io/301-M1-L1/
Recorded finding	The framework is sound and the scenarios are well written, but choosing the middle option nine times without reading scores a perfect 45 of 45, so the build teaches the opposite of its own lesson.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

BOW Sports Capital: Pregame

Canonical ID	bsc-pregame-cap-simulator
Track / slot	Other / cross-track • 101/201 · M1 cap onboarding
Current system	pre-course • active · EXPERIMENTAL · broken • curation HOLD
Mechanic / verbs	business-operator, allocation-under-constraint • operate, invest, allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence unknown

Source	braydenokley13-ux/BSC-pre-course • https://braydenokley13-ux.github.io/BSC-pre-course/
Recorded finding	The most polished cap-management concept in the account, and unplayable: pressing the Continue button it offers after the first decision of week 1 blanks the application, reproducibly.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

Delivery Empire — Economic Strategy Game

Canonical ID	delivery-empire-strategy
Track / slot	101 • Not canonically mapped in the source curriculum
Current system	track-101 • active · PLAYABLE · healthy • curation EXPERIMENTAL
Mechanic / verbs	business-operator, market-and-pricing • operate, invest, price, buy, sell
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/101-M3-ECON • https://braydenokley13-ux.github.io/101-M3-ECON/
Recorded finding	The applied economics content is strong; the interaction is twenty-five multiple-choice questions bookended by two decisions, and a runtime error fires on quick clicking.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

Gauntlet L1: Market Master

Canonical ID	gauntlet-l1-market-master
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	gauntlet • active · EXPERIMENTAL · needs-attention • curation NONE
Mechanic / verbs	market-and-pricing, business-operator • price, buy, sell, operate, invest
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/GAUNTLET • https://braydenokley13-ux.github.io/GAUNTLET/gauntlet-l1/
Recorded finding	Reachable and playable, but its results-submission endpoint is still the placeholder string, so nothing is ever recorded.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

Gauntlet L3: Economic Policy Simulator

Canonical ID	gauntlet-l3-economic-policy-simulator
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum

Current system	gauntlet • active · PLAYABLE · needs-attention • curation EXPERIMENTAL
Mechanic / verbs	allocation-under-constraint, crisis-management • allocate, build, prioritize, respond
Delivery / evidence	individual • facilitator any-instructor • evidence completion
Source	braydenokley13-ux/GAUNTLET • https://braydenokley13-ux.github.io/GAUNTLET/gauntlet-l3/
Recorded finding	Genuine macro tradeoffs across eight quarters with random shocks, but a successful run never resolves: Quarter 8 resubmits forever while a failed run gets a full report.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

Model Risk & False Confidence

Canonical ID	model-risk-false-confidence
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation EXPERIMENTAL
Mechanic / verbs	signal-vs-noise • analyze, forecast
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/301-M4-L3 • https://braydenokley13-ux.github.io/301-M4-L3/
Recorded finding	Sharp, well-researched writing on model risk with no accumulating state behind it. The strongest candidate in the account for turning a quiz into a decision engine.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

NFL System Stress Test

Canonical ID	nfl-system-stress-test
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · needs-attention • curation EXPERIMENTAL
Mechanic / verbs	signal-vs-noise, crisis-management • analyze, forecast, prioritize, respond
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/301-M2-L3 • https://braydenokley13-ux.github.io/301-M2-L3/
Recorded finding	Real Mode plays well and resists guessing, but Tutorial Mode — one of the two routes the entry screen offers — throws on its first question and cannot be completed.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

The Curve Room 2.0 — Salary Cap Simulation

Canonical ID	curve-room-2
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation EXPERIMENTAL
Mechanic / verbs	allocation-under-constraint • allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/301-M1-L2 • https://braydenokley13-ux.github.io/301-M1-L2/
Recorded finding	Six teams across three sports and a real ideal-curve concept, but on a win-now team always spending the most beats the deliberate curve the closing text teaches.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

The League in a Box

Canonical ID	the-league-in-a-box
Track / slot	101 • 101 · M2 L3 / future system design
Current system	track-101 • active · PLAYABLE · healthy • curation REBUILD
Mechanic / verbs	allocation-under-constraint • allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/league-in-a-box • https://braydenokley13-ux.github.io/league-in-a-box/
Recorded finding	The most ambitious concept in the account — design a league and see who benefits — with nothing scarce, no score, and copy written for policy analysts rather than the grades 5 to 8 it targets.
V5 decision	REFOUND. Write a new simulation contract around the concept; current implementation is not the V5 base.

A.4 — MERGE

16 records in this decision class.

Analytics Lab

Canonical ID	analytics-lab
Track / slot	Other / cross-track • 201 · M3 L2
Current system	analytics-lab • active · PLAYABLE · needs-attention • curation EXPERIMENTAL
Mechanic / verbs	signal-vs-noise • analyze, forecast
Delivery / evidence	individual • facilitator any-instructor • evidence structured-assessment
Source	braydenokley13-ux/BSC-BUILDANALYTIC • https://braydenokley13-ux.github.io/BSC-BUILDANALYTIC/track201/
Recorded finding	The weighting tool works and the leaderboard genuinely reorders, but every chart renders empty, the report throws, and selecting a sport fires 24 errors.
V5 decision	MERGE. Merge with Stat Inventor into the Metric Builder / Model Room line.

BOW Sports Capital: Pre-Course

Canonical ID	bsc-pre-course-front-office
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	pre-course • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	allocation-under-constraint, negotiation, shock-and-adapt • allocate, build, negotiate, respond, revise
Delivery / evidence	teams • facilitator bow-certified-instructor • evidence structured-assessment
Source	braydenokley13-ux/BSC-pre-course • No verified live URL
Recorded finding	Requires a deployed Postgres instance; no live URL found. GitHub Pages serves the unrelated standalone game in this repo, not this app.
V5 decision	MERGE. Merge useful onboarding into the shared Economics join and role tutorial.

Decision Room (memo exercise)

Canonical ID	decision-room-memo
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	signal-vs-noise • analyze, forecast
Delivery / evidence	individual • facilitator bow-certified-instructor • evidence scored-outcomes
Source	braydenokley13-ux/301-M4-L1 • No verified live URL

Recorded finding	Not verifiable without the bound Google Sheet, which is not in the repo.
V5 decision	MERGE. Merge the memo evidence into Decision Room Simulator / Model Room.

DeliverEmpire — Build Your Food Delivery Dynasty

Canonical ID	deliverempire
Track / slot	101 • Not canonically mapped in the source curriculum
Current system	track-101 • active · PLAYABLE · healthy • curation EXPERIMENTAL
Mechanic / verbs	business-operator, market-and-pricing • operate, invest, price, buy, sell
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/101-M2-ECON • https://braydenokley13-ux.github.io/101-M2-ECON/
Recorded finding	Real tradeoffs underneath a long quiz, with raw HTML entities printed on screen and a coach who congratulates a score of zero.
V5 decision	MERGE. Merge with Delivery Empire; preserve only the stronger economics cases.

Front Office Challenge

Canonical ID	front-office-challenge
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	sports-agency • active · PLAYABLE · needs-attention • curation NONE
Mechanic / verbs	allocation-under-constraint, shock-and-adapt, negotiation, business-operator • allocate, build, respond, revise, negotiate, operate, invest
Delivery / evidence	teams • facilitator founder-required • evidence educator-report
Source	braydenokley13-ux/bow-finlit • No verified live URL
Recorded finding	Deploy configuration is complete but neither Supabase migration has been applied to a real project, so the live path has never run.
V5 decision	MERGE. Merge its role and data ambitions into the new live Economics product; do not revive the unfinished platform separately.

GM Trade Challenge — The 5-Minute Decision Maker

Canonical ID	gm-trade-challenge
Track / slot	201 • 201 · M4 L3
Current system	track-201 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	trade-and-exchange • trade
Delivery / evidence	individual • facilitator any-instructor • evidence none

Source	braydenokley13-ux/201-M4-L3 • https://braydenokley13-ux.github.io/201-M4-L3/
Recorded finding	Explicit EV, surplus and window numbers with real information asymmetry at higher tiers and a tightening clock. Divergent endings confirmed.
V5 decision	MERGE. Merge decision cases into Draft Asset Market / Trade War Room.

Luxury Tax in Action — Scenario Simulation

Canonical ID	luxury-tax-in-action
Track / slot	201 • 201 · M1 L2
Current system	track-201 • superseded · EXPERIMENTAL · healthy • curation NONE
Mechanic / verbs	allocation-under-constraint • allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/201-M1-L2-Luxury-Tax- • https://braydenokley13-ux.github.io/201-M1-L2-Luxury-Tax-/
Recorded finding	GitHub Pages returned 200 on 2026-08-14. Reachability only.
V5 decision	MERGE. Merge into Cap Crash Control Room and preserve the tax scenario data.

Reputation & the Long Game

Canonical ID	reputation-and-the-long-game
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · EXPERIMENTAL · broken • curation NONE
Mechanic / verbs	negotiation • negotiate
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/301-M3-L3 • No verified live URL
Recorded finding	The workbook states its own automation is not built. Nothing scores, and the results ranges have nothing to populate them.
V5 decision	MERGE. Merge the repeated-game concept into Deal Dynasty / negotiation kernel.

Signal Engine — Sports Agent Negotiation Sim

Canonical ID	signal-engine
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation NONE
Mechanic / verbs	negotiation • negotiate

Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/301-M3-L2 • https://braydenokley13-ux.github.io/301-M3-L2/
Recorded finding	GitHub Pages returned 200 on 2026-08-14. Reachability only.
V5 decision	MERGE. Merge signaling content into the League & Contract Deal Room line after verification.

Sports Analytics Team Builder

Canonical ID	sports-analytics-team-builder
Track / slot	201 • 201 · M3 L3
Current system	track-201 • active · PLAYABLE · healthy • curation REBUILD
Mechanic / verbs	draft-and-selection, allocation-under-constraint • evaluate, select, allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/201-M3-L2 • https://braydenokley13-ux.github.io/201-M3-L2/
Recorded finding	A good premise defeated by its data: every player in the pool clears the thresholds alone, so any legal pair passes and minimum spend equals maximum spend.
V5 decision	MERGE. Merge the organizational premise into Analytics Organization Draft.

Stat Inventor

Canonical ID	stat-inventor
Track / slot	Other / cross-track • 101 · M3 L3 / 201 · M3 L2
Current system	analytics-lab • active · PLAYABLE · needs-attention • curation EXPERIMENTAL
Mechanic / verbs	signal-vs-noise • analyze, forecast
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/BSC-BUILDANALYTIC • https://braydenokley13-ux.github.io/BSC-BUILDANALYTIC/track101/
Recorded finding	A sound metric-construction lab whose charting half does not run at all, and whose phone layout silently drops a data column.
V5 decision	MERGE. Merge with Analytics Lab into one working Metric Builder.

The Asset Everyone Wants

Canonical ID	the-asset-everyone-wants
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bow-website • active · PLAYABLE · healthy • curation RECOMMENDED

Mechanic / verbs	trade-and-exchange • trade
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/BOW-WEBSITE • https://bowsportscapital.com/simulation
Recorded finding	Three chained trade decisions with deltas shown before commitment and a strong closing report. The ending archetype is too coarse to distinguish opposite strategies.
V5 decision	MERGE. Merge the strong chained trade choices into Draft Asset Market.

The Front Office

Canonical ID	the-front-office
Track / slot	201 • Not canonically mapped in the source curriculum
Current system	track-201 • active · PLAYABLE · healthy • curation NONE
Mechanic / verbs	business-operator, trade-and-exchange, crisis-management • operate, invest, trade, prioritize, respond
Delivery / evidence	individual • facilitator any-instructor • evidence educator-report
Source	braydenokley13-ux/BSC-201-Capstone • https://braydenokley13-ux.github.io/BSC-201-Capstone/
Recorded finding	GitHub Pages returned 200 on 2026-08-14. Reachability only.
V5 decision	MERGE. Merge any distinctive cases into Front Office Dashboard; retire duplicate shell.

The GM's Model

Canonical ID	the-gms-model
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · EXPERIMENTAL · needs-attention • curation NONE
Mechanic / verbs	signal-vs-noise • analyze, forecast
Delivery / evidence	individual • facilitator any-instructor • evidence completion
Source	braydenokley13-ux/scout-model • https://braydenokley13-ux.github.io/scout-model/
Recorded finding	GitHub Pages returned 200 on 2026-08-14, but the repository is in poor order — the same application is duplicated across four files and the CI workflow file contains HTML instead of YAML.
V5 decision	MERGE. Merge model disagreement into Model Room; do not preserve the duplicated repository structure.

The Rookie Deal

Canonical ID	the-rookie-deal
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum

Current system	bonus-gm-sims • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	negotiation • negotiate
Delivery / evidence	individual • facilitator any-instructor • evidence completion
Source	braydenokley13-ux/BSC-anythingelse • No verified live URL
Recorded finding	No GitHub Pages deployment (404) and no other live URL found. Requires a build to run.
V5 decision	MERGE. Merge any viable contract case into Contract Table / Agent Simulator.

WAR ROOM — NBA Decision Simulator

Canonical ID	war-room-nba-decision-simulator
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation NONE
Mechanic / verbs	signal-vs-noise • analyze, forecast
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/301-M4-L1 • https://braydenokley13-ux.github.io/301-M4-L1/
Recorded finding	GitHub Pages returned 200 on 2026-08-14. Reachability only.
V5 decision	MERGE. Merge into Front Office Dashboard / Model Room after source verification.

A.5 — SUPPORTING ACTIVITY

4 records in this decision class.

Bow Sports Capital — Creative Business Simulation

Canonical ID	creative-business-simulation
Track / slot	201 • Not canonically mapped in the source curriculum
Current system	track-201 • active · PLAYABLE · healthy • curation EXPERIMENTAL
Mechanic / verbs	business-operator • operate, invest
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/201-M2-ECON • https://braydenokley13-ux.github.io/201-M2-ECON/
Recorded finding	Correctly applied economics chapter by chapter, but no meter, score or resource exists, so the strategic choices only gate which quiz comes next.
V5 decision	SUPPORTING ACTIVITY. Keep as a case, diagnostic, reflection, or short practice; remove flagship-simulation claims.

Bow Sports Empire

Canonical ID	bow-sports-empire
Track / slot	201 • Not canonically mapped in the source curriculum
Current system	track-201 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	business-operator, investment-and-portfolio • operate, invest, allocate
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/201-M3-ECON • https://braydenokley13-ux.github.io/201-M3-ECON/
Recorded finding	The best-looking simulation in the account, and genuinely responsive to choices. The final grade currently ignores the four stats the student spends the game managing.
V5 decision	SUPPORTING ACTIVITY. Keep as a case, diagnostic, reflection, or short practice; remove flagship-simulation claims.

Business Economics Challenge

Canonical ID	business-economics-challenge
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	business-operator • operate, invest
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes

Source	braydenokley13-ux/301-M2-ECON • https://braydenokley13-ux.github.io/301-M2-ECON/
Recorded finding	Rigorous timed economics with the best closing diagnostic in the account — per-concept mastery and a study plan naming what to review. Honestly a quiz rather than a strategy space, and worth using as one.
V5 decision	SUPPORTING ACTIVITY. Keep as a case, diagnostic, reflection, or short practice; remove flagship-simulation claims.

Process vs Results Lab — Football Edition

Canonical ID	process-vs-results-lab
Track / slot	101 • 101 · M4 L1
Current system	track-101 • active · PLAYABLE · healthy • curation RECOMMENDED
Mechanic / verbs	signal-vs-noise • analyze, forecast
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/101-M3-L3 • https://braydenokley13-ux.github.io/101-M3-L3/
Recorded finding	Reaches its first decision with no reading screen at all, has no dominant quadrant, and makes one idea land cleanly. Deliberately narrow.
V5 decision	SUPPORTING ACTIVITY. Keep as a case, diagnostic, reflection, or short practice; remove flagship-simulation claims.

A.6 — ARCHIVE

21 records in this decision class.

BOW Business Strategy Simulator — CandleCart.com

Canonical ID	candlecart-business-strategy
Track / slot	201 • Not canonically mapped in the source curriculum
Current system	track-201 • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	business-operator, market-and-pricing • operate, invest, price, buy, sell
Delivery / evidence	individual • facilitator bow-certified-instructor • evidence educator-report
Source	braydenokley13-ux/201-M1-ECON • No verified live URL
Recorded finding	No GitHub Pages deployment (404). Not verifiable without the bound Google Sheet.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

BOW Sports Capital Zoom Game

Canonical ID	bow-zoom-discovery-game
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	pre-course • superseded · EXPERIMENTAL · needs-attention • curation NONE
Mechanic / verbs	allocation-under-constraint • allocate, build
Delivery / evidence	teams • facilitator bow-certified-instructor • evidence educator-report
Source	braydenokley13-ux/101-pre-course • No verified live URL
Recorded finding	No deployment found; flat-file storage is documented by its own README as unsuitable for production use.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Endorsement Empire

Canonical ID	endorsement-empire
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bonus-gm-sims • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	investment-and-portfolio • invest, allocate
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/BSC-anythingelse • No verified live URL
Recorded finding	No deployment found for this repo (GitHub Pages 404).

V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Fill My Building

Canonical ID	fill-my-building
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bonus-gm-sims • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	market-and-pricing • price, buy, sell
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/BSC-anythingelse • No verified live URL
Recorded finding	No deployment found for this repo (GitHub Pages 404).
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Front Office City — District A

Canonical ID	front-office-city-district-a
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	front-office-city • active · EXPERIMENTAL · needs-attention • curation NONE
Mechanic / verbs	world-exploration, allocation-under-constraint • explore, choose, allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence choices-recorded
Source	braydenokley13-ux/M1-201-FINAL • No verified live URL
Recorded finding	No GitHub Pages deployment (404) and no confirmed live instance. Requires a build and a signed-key deployment to run.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Front Office City: NFL Capital Run

Canonical ID	front-office-city-nfl-capital-run
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	front-office-city • active · PLAYABLE · needs-attention • curation NONE
Mechanic / verbs	world-exploration, negotiation • explore, choose, negotiate
Delivery / evidence	individual • facilitator any-instructor • evidence educator-report
Source	braydenokley13-ux/M2-201-FINAL • No verified live URL
Recorded finding	No GitHub Pages deployment (404); CI runs tests but does not publish. No live instance found.

V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Front Office — Eastfield Eagles

Canonical ID	eastfield-eagles
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bow-website • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	allocation-under-constraint • allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/BOW-WEBSITE • No verified live URL
Recorded finding	The route exists in a deployed site, but it is gated behind a student account and prior module completion, so the experience itself could not be reached and verified.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Ground Zero

Canonical ID	ground-zero
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bonus-gm-sims • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	draft-and-selection, allocation-under-constraint • evaluate, select, allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence completion
Source	braydenokley13-ux/BSC-anythingelse • No verified live URL
Recorded finding	No GitHub Pages deployment (404) and no other live URL found. Requires a build to run.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Highway World — Challenge Mode: Game Night Pricing Crisis

Canonical ID	highway-world-challenge-mode
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	highway-world • active · EXPERIMENTAL · needs-attention • curation NONE
Mechanic / verbs	market-and-pricing, crisis-management • price, buy, sell, prioritize, respond
Delivery / evidence	individual • facilitator any-instructor • evidence facilitator-observation
Source	braydenokley13-ux/BSC-HIGHWAY-WORLD • No verified live URL
Recorded finding	Shares the campaign's lack of any deployment.

V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Highway World — Episode 1: Origin District

Canonical ID	highway-world-episode-1
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	highway-world • active · PLAYABLE · needs-attention • curation NONE
Mechanic / verbs	world-exploration, allocation-under-constraint, negotiation, market-and-pricing, draft-and-selection • explore, choose, allocate, build, negotiate, price, buy, sell, evaluate, select
Delivery / evidence	individual • facilitator any-instructor • evidence facilitator-observation
Source	braydenokley13-ux/BSC-HIGHWAY-WORLD • No verified live URL
Recorded finding	No deployment of any kind: no hosting config and no publish step in CI. An instructor would have to clone the repository and run a dev server.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Module 4 Econ Final — Track 101

Canonical ID	module-4-econ-final-101
Track / slot	101 • Not canonically mapped in the source curriculum
Current system	track-101 • active · EXPERIMENTAL · unknown • curation NONE
Mechanic / verbs	not classified •
Delivery / evidence	individual • facilitator bow-certified-instructor • evidence choices-recorded
Source	braydenokley13-ux/101-M4-ECON • No verified live URL
Recorded finding	No GitHub Pages deployment (404). Cannot be verified without the bound Google Sheet, which is not in the repo.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Plan Under Pressure

Canonical ID	plan-under-pressure
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	decision-challenges • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	allocation-under-constraint, shock-and-adapt • allocate, build, respond, revise
Delivery / evidence	individual • facilitator any-instructor • evidence structured-assessment
Source	braydenokley13-ux/bow-decision-challenges • No verified live URL

Recorded finding	Configured to deploy to Vercel but no live URL appears anywhere in the repo, so reachability could not be probed.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Sandbox Impact Model

Canonical ID	bow-universe-sandbox
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bow-universe • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	market-and-pricing, allocation-under-constraint • price, buy, sell, allocate, build
Delivery / evidence	individual • facilitator any-instructor • evidence structured-assessment
Source	braydenokley13-ux/bow-universe • No verified live URL
Recorded finding	No live URL found; requires a Postgres deployment to run.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Simulation Room — Westbrook Wolves

Canonical ID	westbrook-wolves
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bow-website • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	allocation-under-constraint, business-operator • allocate, build, operate, invest
Delivery / evidence	individual • facilitator any-instructor • evidence scored-outcomes
Source	braydenokley13-ux/BOW-WEBSITE • No verified live URL
Recorded finding	The route exists in a deployed site, but it is gated behind a student account and prior module completion, so the experience itself could not be reached and verified.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Startup Tycoon: Data-Driven Decisions

Canonical ID	startup-tycoon
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · PLAYABLE · broken • curation NONE
Mechanic / verbs	signal-vs-noise, investment-and-portfolio • analyze, forecast, invest, allocate
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/301-M4-ECON • https://braydenokley13-ux.github.io/301-M4-ECON/

Recorded finding	GitHub Pages returned 404 "Site not found" on 2026-08-16. The repository exists but has no Pages deployment, so the recorded live URL opens nothing. It answered 200 on 2026-08-14, so this is a regression rather than a link that was never right.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Structural Leverage SIM

Canonical ID	structural-leverage-sim
Track / slot	301 • Not canonically mapped in the source curriculum
Current system	track-301 • active · EXPERIMENTAL · broken • curation NONE
Mechanic / verbs	negotiation • negotiate
Delivery / evidence	individual • facilitator bow-certified-instructor • evidence scored-outcomes
Source	braydenokley13-ux/301-M3-L1 • No verified live URL
Recorded finding	The browser build has no HTML entry point — the repo's only commit is titled 'Delete index.html' and none exists in the tree. The spreadsheet build appears complete but cannot be verified without the bound sheet.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

Tank Commander

Canonical ID	tank-commander
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bonus-gm-sims • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	signal-vs-noise • analyze, forecast
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/BSC-anythingelse • No verified live URL
Recorded finding	No deployment found for this repo (GitHub Pages 404).
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

The Blockbuster

Canonical ID	the-blockbuster
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bonus-gm-sims • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	negotiation, trade-and-exchange • negotiate, trade
Delivery / evidence	individual • facilitator any-instructor • evidence completion

Source	braydenokley13-ux/BSC-anythingelse • No verified live URL
Recorded finding	No GitHub Pages deployment (404) and no other live URL found. Requires a build to run.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

The Dumpster Fire

Canonical ID	the-dumpster-fire
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bonus-gm-sims • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	allocation-under-constraint, shock-and-adapt • allocate, build, respond, revise
Delivery / evidence	individual • facilitator any-instructor • evidence completion
Source	braydenokley13-ux/BSC-anythingelse • No verified live URL
Recorded finding	No GitHub Pages deployment (404) and no other live URL found. Requires a build to run.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

The Front Office (early build)

Canonical ID	the-front-office-early-build
Track / slot	201 • Not canonically mapped in the source curriculum
Current system	track-201 • superseded · EXPERIMENTAL · healthy • curation NONE
Mechanic / verbs	business-operator • operate, invest
Delivery / evidence	individual • facilitator any-instructor • evidence unknown
Source	braydenokley13-ux/Franchise-Sim • https://braydenokley13-ux.github.io/Franchise-Sim/
Recorded finding	GitHub Pages returned 200 on 2026-08-14. Reachable, but superseded.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

The LeBron Files

Canonical ID	the-lebron-files
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	bonus-gm-sims • active · PLAYABLE · unknown • curation NONE
Mechanic / verbs	business-operator, investment-and-portfolio • operate, invest, allocate
Delivery / evidence	individual • facilitator any-instructor • evidence completion
Source	braydenokley13-ux/BSC-anythingelse • No verified live URL

Recorded finding	No GitHub Pages deployment (404) and no other live URL found. Requires a build to run.
V5 decision	ARCHIVE. Preserve provenance and source links; remove from the active teaching and discovery surface.

A.7 — KILL

2 records in this decision class.

BOW Sports Capital — Final Mastery Simulation

Canonical ID	final-mastery-simulation-201
Track / slot	201 • Not canonically mapped in the source curriculum
Current system	track-201 • active · EXPERIMENTAL · broken • curation NONE
Mechanic / verbs	not classified •
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/201-M4-ECON • No verified live URL
Recorded finding	The file contains 154 lines of constants and zero functions. Opening it in a spreadsheet produces no menu and no runnable activity.
V5 decision	KILL. Remove from active and R&D shelves; retain only the audit record and historical source reference.

Entrepreneurship Lab — Unit Economics

Canonical ID	entrepreneurship-lab-unit-economics
Track / slot	Other / cross-track • Not canonically mapped in the source curriculum
Current system	entrepreneurship-lab • active · EXPERIMENTAL · broken • curation NONE
Mechanic / verbs	business-operator, market-and-pricing • operate, invest, price, buy, sell
Delivery / evidence	individual • facilitator any-instructor • evidence none
Source	braydenokley13-ux/entrepuernurship • No verified live URL
Recorded finding	Deployed and dead. The page is live at https://braydenokley13-ux.github.io/entrepuernurship/entrepeurneurship/ and returns HTTP 200 with its full UI rendered, but a syntax error (Unexpected token ':') stops every script: Start Run and all four strategy presets were clicked in a browser on 2026-08-16 and changed nothing on the page. index.html never includes state.js.
V5 decision	KILL. Remove from active and R&D shelves; retain only the audit record and historical source reference.

APPENDIX B

Reusable V5 contracts

The minimum governance needed to build consistently without turning BOW into a bureaucracy.

B.1 Lesson blueprint

Identity	Track; module; lesson; title; owner; version; target age; duration.
Learning	Concept IDs; age-banded objective; misconception; prerequisite; model boundary.
Experience	Lesson form; hook; prediction; teaching; student role; grouping; device plan; first decision.
System	Simulation/activity version; agents; incentives; constraints; information; actions; feedback; dynamics.
Facilitation	Controls; freeze points; questions; timing branches; room-energy protocol; recovery/fallback.
Data	Required events; metrics; graphs; comparison; invalid-run criteria; retention.
Debrief	Notice; theory; boundary; counterfactual; explanation; unresolved question.
Validation	Economics review; grade-band usability; technical test; live receipts; next decision.

B.2 Simulation contract

Purpose	The one relationship the system must make visible and the learning job it does in the lesson.
Agents	Roles and who/what each represents.
Incentives	What each agent tries to achieve; whether incentives conflict or align.
Constraints	Budgets, rules, time, capacity, legal actions, and what cannot be changed.
Information	Public, private, estimated, hidden, and when revelation is allowed.
Actions	The player verbs and validated parameters; first consequential action.

Transitions	Authoritative state changes, timing, seeded randomness, counterpart/system response.
Metrics	Declared class metrics, units, formulas, uncertainty, and comparison-set rule.
Controls	Start/pause/freeze/end, valid shocks, reveal, rerun, recovery, and restore semantics.
Debrief	Prediction, reveal, graph, mechanism, model boundary, counterfactual, questions.
Quality	Accessibility, privacy, performance, failure state, fallback, owner, health, live receipts.

B.3 Experience truth states

State	May claim	May not claim
Discovered	A source exists.	It runs or teaches the tagged concept.
Reachable	The launch URL loads.	The critical path completes.
Playable	An adult completed the critical path.	Students understand or teachers can facilitate it.
Ramaz-tested	This version ran in the named session; receipt exists.	It transfers or produces durable learning.
Ramaz-ready	Current build, guide, recovery, data, and debrief are usable by Brayden.	A non-founder can run it.
Transfer-ready	Named transfer test passed.	It generalizes to all schools or motifs.
Evidence-bearing	A defined evidence contract supports a bounded reading.	Mastery, causality, or durability beyond that reading.

APPENDIX C

Future curriculum opportunities

Keep the current Track 101/201 sequence stable while prototyping the missing economic systems separately.

Opportunity	Signature experience	Why it belongs later
Market formation	Your Market: private values/costs, bids/asks, price convergence, surplus, tax or price rule rerun.	Most direct class-generated data signature and real-time platform stress test.
Specialization & trade	Teams produce under different productivity, then open exchange and compare total output.	Makes comparative advantage social and visible without advanced math.
Public goods	Groups allocate private resources to a shared outcome with changing contribution rules.	Adds collective-action mechanic absent from current sports portfolio.
Commons	Students extract from a renewable shared resource and design an institution after depletion risk appears.	Creates surprising aggregate outcomes and rule-design discussion.
Competition & market power	Firms set price/quantity under entry, differentiation, or monopoly rules.	Expands beyond roster allocation and ties strategic interaction to market structure.
Externalities & policy	Production generates private value and public cost; class tests tax, cap, disclosure, or rights.	Supports counterfactual policy design with explicit model boundaries.
Coordination & network effects	Students choose incompatible standards and experience lock-in/switching costs.	Adds system dynamics and strategic expectations.
Labor market & information	Workers and firms match with private skill/productivity signals and contracts.	Connects bargaining, signaling, and human-capital choices.

APPENDIX D

Source and evidence ledger

The sources V5 used, the role each source played, and the limits attached to it.

Primary BOW sources

BOW Model V4: https://docs.google.com/document/d/1G-7HY_c43nl3sbUC2le7Wn9Je3Hx97qy5vuFbiIBN0o/edit?tab=t.0 — Strategic baseline: concept registry, product separation, module doctrine, challenge/evidence boundaries, technology and governance.

Economics curriculum PDF — BOW Sports Capital Podcast-4.pdf (315 pages; supplied from the user's Downloads folder). Used to reconstruct Track 101/201 modules, lessons, scripts, activities, examples, and source contradictions.

Simulation Library: https://github.com/braydenokley13-ux/sim-library — Canonical 76-record audit, health, curation, run resources, provenance, and recorded playtest findings.

Decision Challenges definition package — local project files 06, 15, 17, and 22. Used for the current Challenge thesis, dual content/evidence architecture, action-construct fidelity, portfolio doctrine, and near-term Financial Literacy boundary.

External live-classroom benchmarks

MobLab classroom games: https://www.moblab.com/products/classroom — Benchmark for instructor setup, real-time launch/monitoring, device participation, and game-result teaching.

MobLab modules: https://www.moblab.com/modules — Benchmark for connecting game, reflection, discussion, and instructor guides.

oTree live pages: https://otree.readthedocs.io/en/master/live.html — Reference for real-time messages, bids, and multiplayer interaction patterns.

oTree rooms: https://otree.readthedocs.io/en/master/rooms.html — Reference for short links and waiting-room session entry.

oTree wait pages: https://otree.readthedocs.io/en/latest/multiplayer/waitpages.html — Reference for grouping, arrival, dropouts, bots, and slow-participant handling.

EconPort double auction handbook: https://econport.gsu.edu/content/handbook/industrialorg/marketin/double.html — Reference for classroom double-auction convergence with private values and costs.

Evidence limits

- The curriculum PDF is not a clean final curriculum and contains outdated or contradictory sports facts. Durable V5 architecture should not depend on those facts.
- The Simulation Library's recorded adult browser tests are valuable technical and design evidence; they are not student learning or facilitator-transfer evidence.
- External products are principle benchmarks, not templates to copy. BOW's distinctive opportunity is the integrated live lesson, class-generated data, counterfactual rerun, and subject-truth contract.
- Any stronger claim—learning gain, transfer, durability, broad school fit, or motif comparability—requires an explicit study or validation set.

APPENDIX E

Claude Code Platform + Economics Gauntlet

The staged implementation prompt for connecting the BOW control plane, Simulation Library, and first refounded Economics runtime.

BEGIN PROMPT

Copy from this heading through END PROMPT into Claude Code with access to BOW-WEBSITE, sim-library, and the approved Economics runtime repository. Work repository by repository with explicit ownership and compatibility contracts. This prompt authorizes a staged refoundation, not wholesale deletion or migration.

Mission

You are the implementation lead for the BOW control plane and next-generation BOW Economics. BOW-WEBSITE is the canonical control plane and public entry. sim-library remains the canonical product registry and governance source. The approved Economics runtime owns live student, facilitator, display, economy-state, and debrief behavior. Your job is to connect those boundaries safely, then deliver a Ramaz-ready live Economics system with versioned contracts, class-generated data, recoverable sessions, truthful usage, and three complete prototype vertical slices.

Do not treat this as a request to restyle the current library, rewrite all 76 simulations, merge every repository, rebuild Financial Literacy, rebuild Decision Challenges, redesign slides, or construct a second LMS. Economics is the first implementation. The existing registry, BOW-WEBSITE operational data, and their provenance must survive.

Read before changing anything

1. Read every repository instruction file, especially AGENTS.md, README, CATALOG, data schema, validation scripts, deployment configuration, and test commands.
2. Inspect the current worktree and preserve unrelated or user-authored changes. Do not reset, delete, or reformat broadly.

3. Inventory the actual stack, package manager, routes, data pipeline, and deployment model. Do not assume this prompt's file names already exist.
4. Read the accepted BOW Model V5 definition package and extract the decisions, working specifications, hypotheses, lesson contracts, and promotion gates.
5. Create a new working branch if the environment permits. Do not push, open a pull request, publish, migrate production data, or change external services without explicit authorization.

Non-negotiable product decisions

- Shared BOW core, distinct product contracts. This repository may implement Economics and the Simulation Library; it must not couple Economics to the Decision Challenges runtime.
- This implementation targets Economics × Enrichment at Ramaz. Do not encode enrichment-only assumptions into the Economics concept canon or simulation contracts; a future school-day arm must be able to add standards, transfer, and evidence without rewriting the subject model.
- Economics is live and facilitator-led when interdependence matters. Accounts are off by default; students join with a short code and safe name/alias.
- The three primary views are Player, Facilitator, and Class Display. All three are part of each prototype vertical slice.
- The session engine is authoritative. Clients cannot create resources, change hidden state, or submit actions that violate the simulation contract.
- Randomness is seeded and recorded. Every stochastic outcome used in a debrief can be reproduced.
- Debrief is product. Each prototype declares events, metrics, graphs, comparison rules, model boundaries, and invalid-run criteria.
- Data describes before it explains. The UI must visually distinguish observed result, model expectation, and instructor interpretation.
- Multiplayer is used only where students materially affect one another. Do not turn individual allocation into fake simultaneous multiplayer.
- Accessibility, privacy, reconnect, restore, bot takeover, and fallback are acceptance criteria—not later polish.
- Do not migrate all 76 experiences. Keep legacy entries linkable and versioned while the new flagship contracts are proven.

Definition of the first implementation

Build the smallest control-plane integration and live-economy runtime capable of supporting three prototypes. Use BOW-WEBSITE for identity, organizations, entitlements, catalog/session index, signed launches, usage, and evidence references. Use the Economics runtime for authoritative room state and interaction. Use sim-library for manifest, provenance, health, validation, and promotion. Record every boundary in an architecture decision record. Do not select a real-time vendor until you compare deployment constraints, authoritative-room support, reconnect behavior, operational burden, and cost.

Surface	Required first release
Player	Join code; alias; role/private info; one-screen goal; validated action; immediate feedback; reconnect; keyboard-complete flow.
Facilitator	Create session; assign/group; start; pause; freeze; advance/end; reveal; named shock; rerun; class data; reassign; bot; restore snapshot.
Display	Join/readiness; prediction; public live state; freeze/event; class metrics; graph; comparison; debrief. No private or identified data.
Session engine	Authoritative state; role assignment; command validation; timers; snapshots; seeded randomness; event write; deterministic replay; reconnect token.
Simulation kernel	Pure rules separate from UI: agents, incentives, constraints, information, actions, transitions, metrics, model boundary.
Registry	Canonical experience and version record; curriculum mapping; health; owner; run resource; promotion state; live receipts.

Phase 0 — Source truth and contracts

1. Run the repository's existing checks before edits and record the baseline, including any failures that predate this work.
2. Write an ADR that describes the existing architecture, the V5 target, what will be shared, what remains legacy, the chosen application/package boundary, and why no universal BOW runtime is being built.
3. Create machine-readable schemas and human-readable examples for: Concept Contract, Lesson Blueprint, Simulation Contract, Class Data Contract, Debrief Recipe, Experience Version, and Live Session Receipt.
4. Map the three prototypes to Track/module/lesson, concept IDs, model boundaries, student verbs, grouping, facilitator controls, events, metrics, graphs, shocks, fallback, and promotion gate.
5. Add schema validation and fixtures. Validation failures must name the exact field and experience. Existing 76 records must remain readable through an explicit migration/compatibility path; no silent data loss.

6. Produce an audit report of duplicate IDs, missing owners, broken run resources, unmapped lessons, stale verification dates, and records that cannot satisfy the new schema.

PHASE 0 GATE

Do not begin the live runtime until contracts validate, the compatibility plan is explicit, baseline failures are separated from new failures, and the ADR can be reviewed without reading source code.

Phase 1 — Live session shell

1. Implement session creation and short-code joining with ephemeral alias identity. Do not require a student account.
2. Implement authoritative roles, commands, timing, seeded randomness, snapshots, reconnect, duplicate-join handling, facilitator reassignment, bot takeover, and deterministic replay.
3. Implement the shared room-state machine: LOBBY → PREDICTION → BRIEFING → LIVE → PAUSED/FROZEN → CLOSED → DEBRIEF → COMPLETE. Invalid transitions must be rejected server-side.
4. Implement stable facilitator controls and command audit events. PAUSE and FREEZE must have different semantics as defined in V5.
5. Implement display states with privacy-safe aggregate data and a clear observed/model/interpretation distinction.
6. Add a simulated-class harness that can create at least 32 participants, configurable delays, disconnects, invalid actions, and bot strategies.

PHASE 1 GATE

Demonstrate a 32-participant session through join, action, freeze, reconnect, snapshot restore, close, metrics, and replay. No client may mutate authoritative state. The display must not expose private fields.

Phase 2 — Three complete prototype vertical slices

Prototype A — Your Market

- Roles: buyers with private willingness to pay and sellers with private cost.
- Actions: bid, ask, accept/cross according to the selected market mechanism.
- Round 1: no policy change. Round 2: one named condition such as per-unit tax, disclosure rule, or fixed-price constraint; hold the value/cost distribution constant when comparison requires it.

- Metrics: bids, asks, trades, price, quantity, buyer/seller/total surplus, failed trades, convergence, tax revenue if applicable.
- Display: public market state during play; class supply/demand and transaction graph only after close unless the contract explicitly allows strategic visibility.
- Model boundary: simplified classroom double auction; no claim that one class exactly reproduces a competitive market.

Prototype B — Cap Builders

- Track job: 101 M1, with a 201 rule-depth configuration later.
- Actions: build a legal mini-roster under hard-cap, soft-cap, and/or tax/apron rules; the system computes arithmetic.
- State: talent, cost, depth, exceptions, restrictions, and flexibility. A star-heavy path must not dominate by construction.
- Shock: injury, retention need, owner budget, or deadline opportunity selected from pre-authored economically valid events.
- Metrics: legal builds, spend, expected performance range, depth, option value/flexibility, actions lost, adaptation.
- Model boundary: educational roster model; label excluded league rules and dated values.

Prototype C — League Balance Lab

- Track job: 101 M2 L2–L3.
- Roles: teams with different local markets; optional league/player stakeholder roles only if they create real decisions.
- Actions: choose or vote on revenue-sharing and cap institutions, then allocate roster resources under the resulting budgets.
- Comparison: run at least two institution versions with the same market starting data when making a counterfactual comparison.
- Metrics: revenue distribution, payroll distribution, competitive balance, star concentration, stakeholder outcomes, rule votes.
- Model boundary: no rule is labeled universally fair or efficient; the debrief names the objective optimized and outcomes omitted.

PHASE 2 GATE

Each prototype must have all three views, keyboard-complete critical path, automated unit/integration/end-to-end tests, simulated-class load/reconnect tests, seeded replay, debrief recipe, model-truth review, mobile/Chromebook and projector visual QA, and a facilitator fallback. A player-only prototype does not pass.

Phase 3 — Ramaz Alpha package

1. Create a six-lesson Alpha plan: 101 Salary Cap Basics, 101 Small Markets, 101 Why the Draft Isn't a Ranking, 201 How Teams Bend the Cap, 201 Player Economics, and 201 Modeling the Market.
2. For experiences not rebuilt in Phase 2, create a governed legacy launch package with health status, fallback, facilitation guide, and debrief plan. Do not pretend they are on the new runtime.
3. Create facilitator rehearsal mode, demo session data, printable role/fallback cards, projector checks, and a technical incident protocol.
4. Implement the Live Session Receipt and a private operator review workflow. Do not build a gradebook or individual mastery score.
5. Update the Simulation Library shelves and curation status from real evidence: Ramaz-ready, Core curriculum, Supporting activity, R&D bench, Archive.

Required tests

Test family	Minimum coverage
Kernel unit	Valid/invalid actions, resource conservation, legal state, metrics, boundaries, seeded randomness, shock transitions.
Property/invariant	No impossible roster, no negative resource unless declared, surplus formulas, transaction conservation, role privacy.
Session integration	State transitions, timing, freeze, close, reconnect, duplicate join, reassignment, bot, restore, replay.
End-to-end	Teacher creates; students join; prediction; round; reveal; rerun; debrief; receipt on supported viewport/device profiles.
Load/failure	At least 32 simulated participants; delay, disconnect, reconnect, duplicate, malformed action, display loss.
Accessibility	Automated scan plus manual keyboard, focus, screen-reader labels, non-color states, reduced motion, graph alternatives.
Privacy/security	No private fields on display; no student PII in logs; session-code abuse limits; server validation; retention test.
Visual	Student mobile/Chromebook, facilitator laptop, and 16:9 projector screenshots at every major state; no clipped/overlapping text.

Independent gauntlet reviews

For each prototype, create a short review artifact from each lens below. A builder may draft the evidence, but may not self-certify a pass without the cited test or reviewer observation.

Reviewer	Reject when
Economics truth	Incentives, accounting, causal story, or boundary is dishonest or a predetermined conclusion is rigged.
Game/system design	Student verbs are read/click/read; state does not matter; other players are decorative; replay has no strategic meaning.
Track 101 usability	Rules or arithmetic block the decision; role/goal is unclear; consequence cannot be explained.
Track 201 depth	One heuristic or middle option dominates; uncertainty and context are cosmetic.
Live teacher	A single instructor cannot run, freeze, recover, reveal, and debrief while watching the room.
Visual product	The experience is a generic dashboard, form, or worksheet; motion and hierarchy do not reveal consequence.
Architecture	The implementation couples to Challenges, forces one mechanic, hides versioning, or requires migrating legacy content.
Skeptical product	The experience could be replaced by a quiz without losing the learning mechanism.

Repository and change discipline

- Use the repository's package manager and established conventions. Add dependencies only with a documented need and maintenance owner.
- Prefer pure, testable simulation kernels and adapters over conditionals spread through UI components.
- Preserve registry provenance and immutable historical versions. Schema migrations must be explicit, reversible where practical, and tested on all 76 records.
- Do not delete or rename legacy records to make tests pass. Mark archive/kill status through governed fields.
- Do not hardcode current sports cap figures into durable logic. Use versioned scenario data and display an as-of date.
- Do not add AI-generated teacher conclusions or student scores. Any assistance must cite the events/metrics it summarizes and remain optional.
- Keep commits scoped and explain architecture, data, tests, visual work, and migration separately when the workflow supports commits.

Final handoff

When the authorized phases are complete, return a plain-language handoff that begins with what now works. Include:

- The exact phases completed and gates passed or failed.
- A map of the new architecture and how legacy registry records remain compatible.
- How to run tests, start the product, create a session, join as students, open facilitator view, and open class display.
- Screenshots or recordings of every major state for all three prototypes and all three surfaces.
- Test results, simulated-class size, reconnect/restore evidence, accessibility checks, privacy checks, and known limitations.
- The three economist/model-boundary reviews and the eight adversarial review artifacts.
- The updated library curation report without pretending untested experiences are Ramaz-ready.
- The next smallest safe step. Do not call the refoundation complete merely because the build passes.

END PROMPT

Stop here. The implementation prompt is complete.

Closing recommendation

BOW should not choose between being a curriculum company, a simulation company, a consumer company, or an infrastructure company by building four disconnected businesses. It should own a recognizable category—decision learning—then let school products, Decision Challenges, BOW League, partner infrastructure, and creators express that category through one trusted platform.

Economics is the right first proving ground because Brayden can teach live, the room can become the economy, and class-generated data can turn an abstract graph into something students made. Financial Literacy is the strongest recurring school-day revenue wedge because requirements, family relevance, and practical application create durable demand. Decision Challenges provide depth and evidence. BOW League creates persistent consumer identity. Infrastructure and the Exchange become the long-term platform ceiling only after the underlying experiences and contracts are proven.

The company should maximize reach without making children the monetized object. Free flagships create adoption. Schools pay for managed implementation. Sponsors fund access. Families pay for an exceptional optional League. Enterprises pay for customization and engagement. Partners pay for infrastructure and licensing. Qualified creators eventually expand supply and share in the value they create.

THE V5 BET

Build a small number of exceptional decision systems, prove that students learn and return, make them transferable beyond Brayden, and place them behind one identity, entitlement, catalog, session, usage, and evidence control plane. Then let free reach, institutional revenue, sponsored access, consumer membership, enterprise products, infrastructure, and creators compound around the same trusted core. That is the model with both mission integrity and a credible $10B ceiling.